



Visual AI: From Visual Sensing to Visual Intelligence

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Artificial Intelligence

Intelligence



Computer

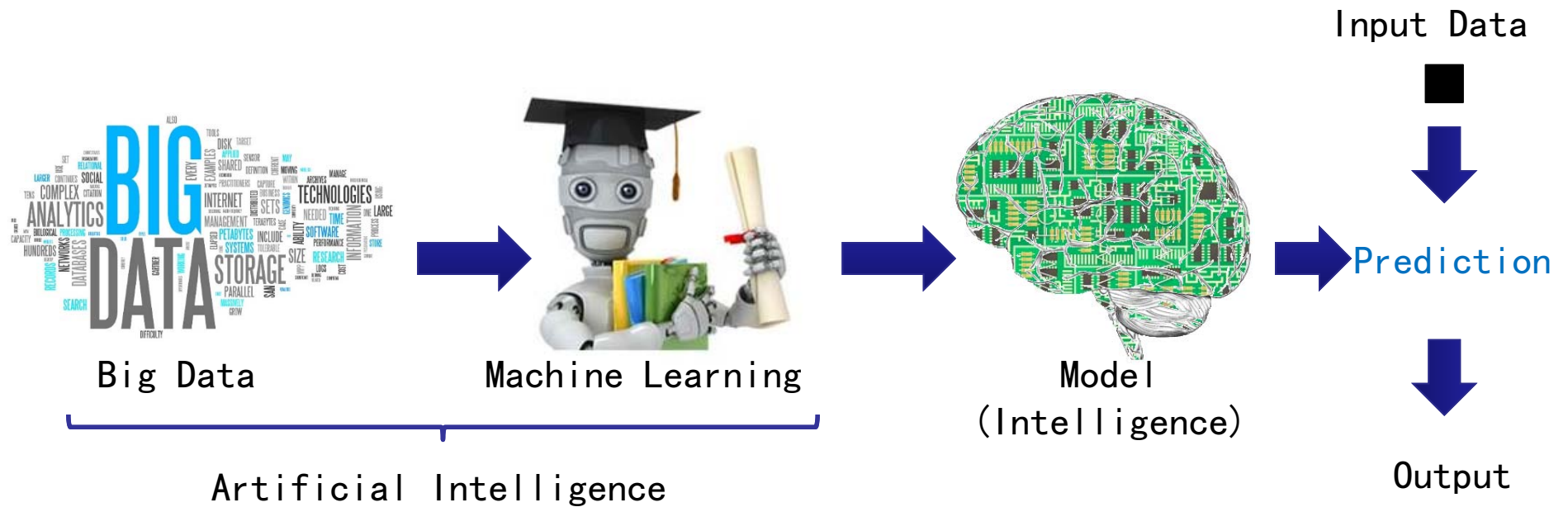
versus

Human

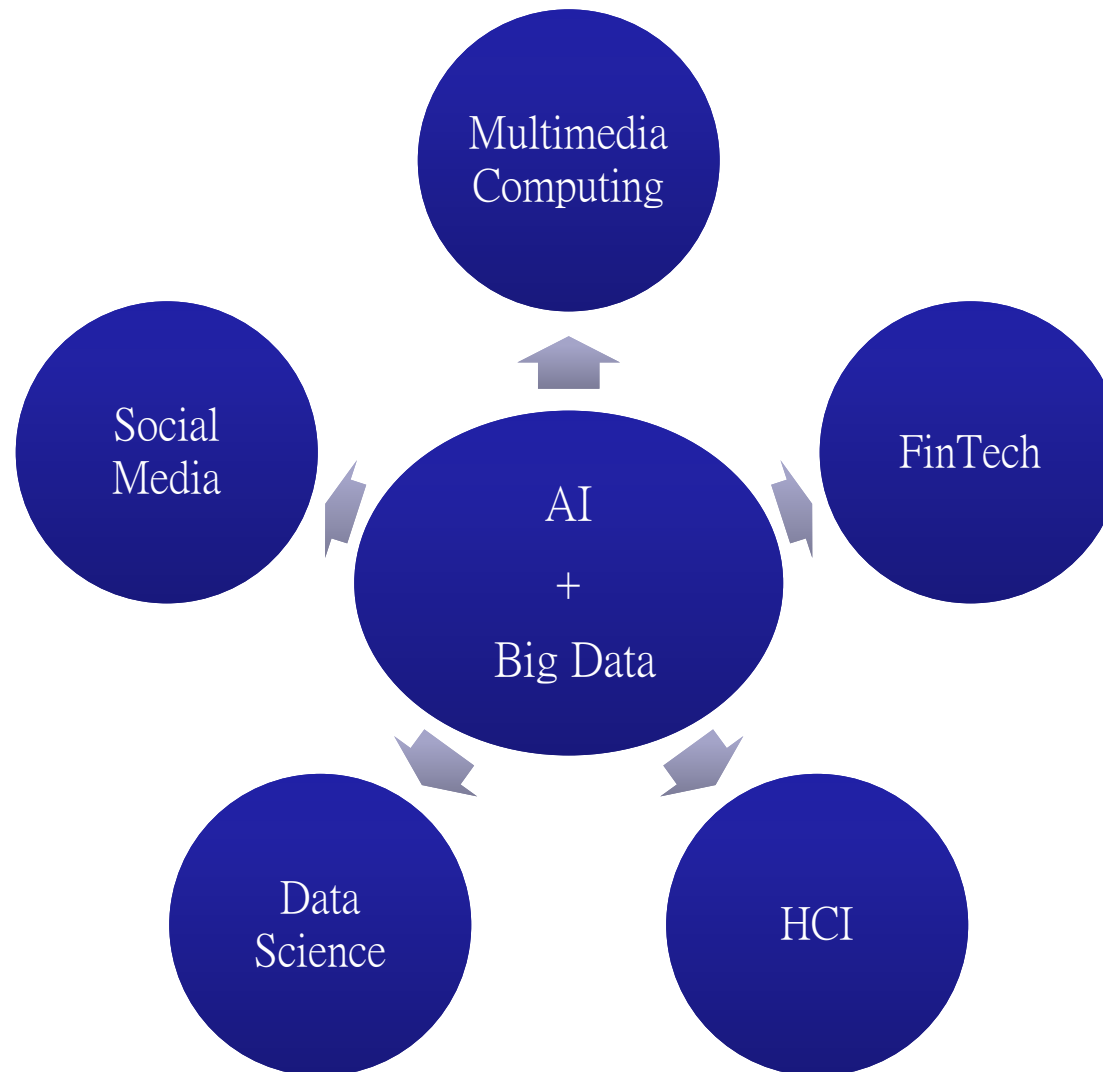


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AI = Big Data+ Machine Learning



Research Area



Multimedia Computing Lab (MCLab)



Welcome to the Multimedia Computing Laboratory (MCLab)!



[A view around the MCLab in Academia Sinica]

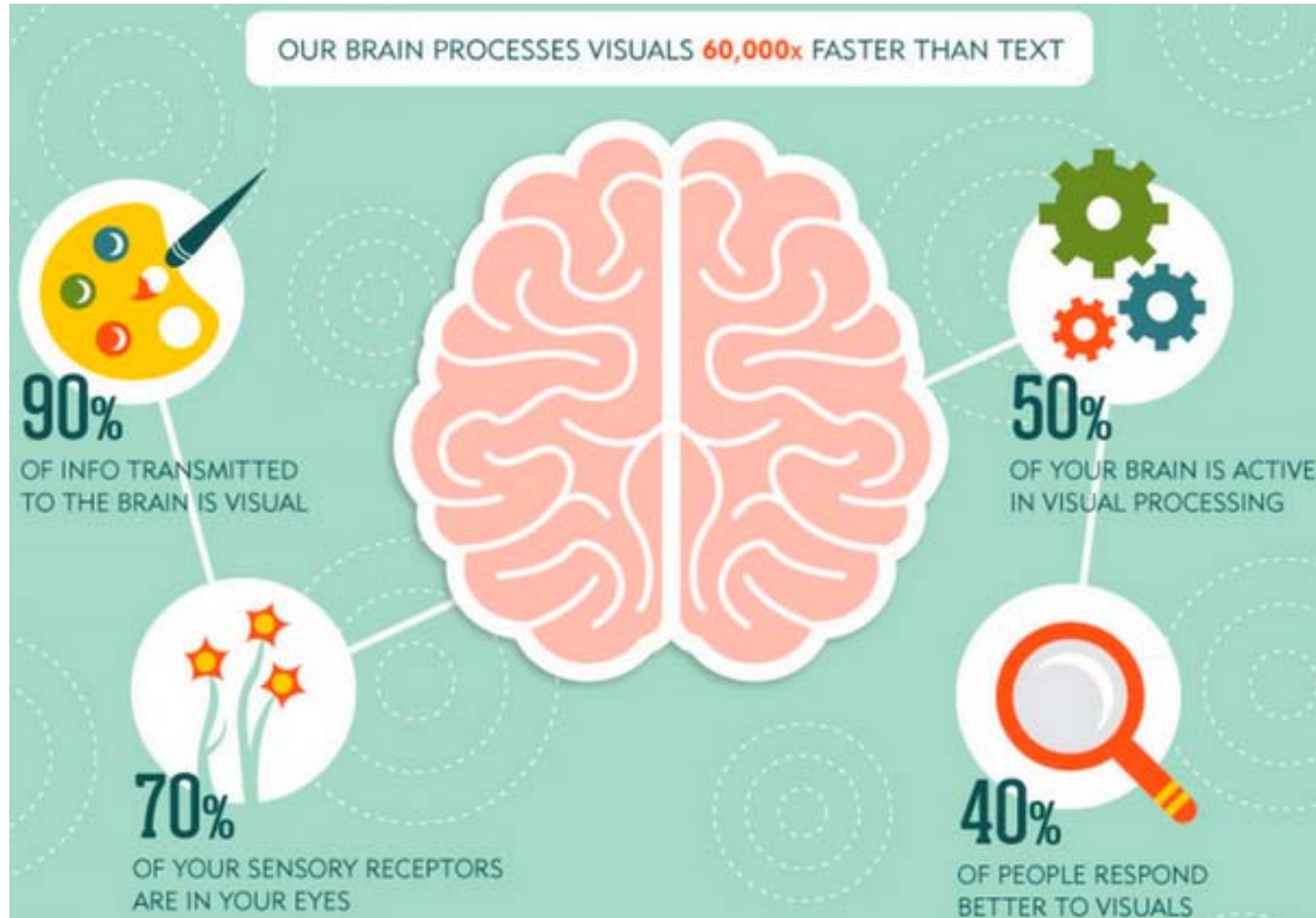


<http://mclab.citi.sinica.edu.tw>



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We live in a visual world...



Computer Vision

*“Connect a television camera to a computer
and get the machine to describe what it sees.”*

- Marvin Minsky (1966), Founder of MIT AI Lab



What is Computer Vision?

- A major branch of Artificial Intelligence
- Vision is the process of discovering from images **what is present in the world**, and **where it is**.

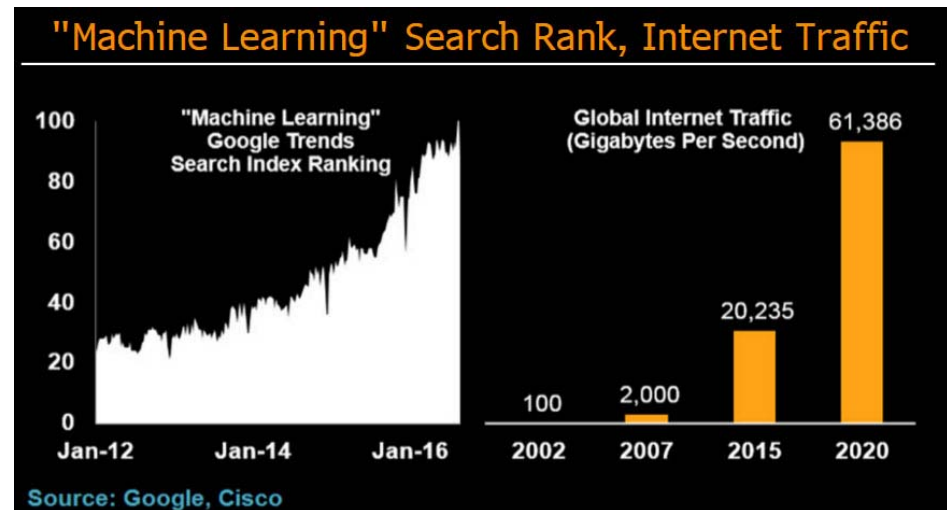
- David Marr, Vision (1982)



What Factors Propelled the AI Advances?

- Sensor technologies and numbers
- Big data
- Computing power
- Human power!

- Key to AI: ABCDE
 - A(lgorithm)
 - B(ig data)
 - C(omputing)
 - D(omain knowledge)
 - E(cosystem)



Selected Publications

- "Robust RGB-D Hand Tracking Using Deep Learning Priors," **IEEE Transactions on Circuits and Systems for Video Technology**, 2017.
- "Learning and Recognition of Clothing Genres from Full-body Images," **IEEE Transactions on Cybernetics**, 2017
- "Fashion World Map: Understanding Cities Through Streetwear Fashion" **MM** 2017.
- "Sequential Prediction of Social Media Popularity with Deep Temporal Context Networks," **IJCAI** 2017.
- "i-Stylist: Finding the Right Dress Through Your Social Networks," **MMM** 2017.
- "Time Matters: Multi-scale Temporalization of Social Media Popularity" **MM** 2016.
- "Unfolding Temporal Dynamics: Predicting Social Media Popularity Using Multi-scale Temporal Decomposition," **AAAI** 2016.
- "UbiShop: Commercial Item Recommendation Using Visual Part Based Object Representation," **Multimedia Tools and Applications**, 2016.
- "A Framework of Enlarging Face Datasets Used for Makeup Face Analysis," **BigMM** 2016.
- "What are the Fashion Trends in New York?" **MM** 2014.



Social Signals





There are

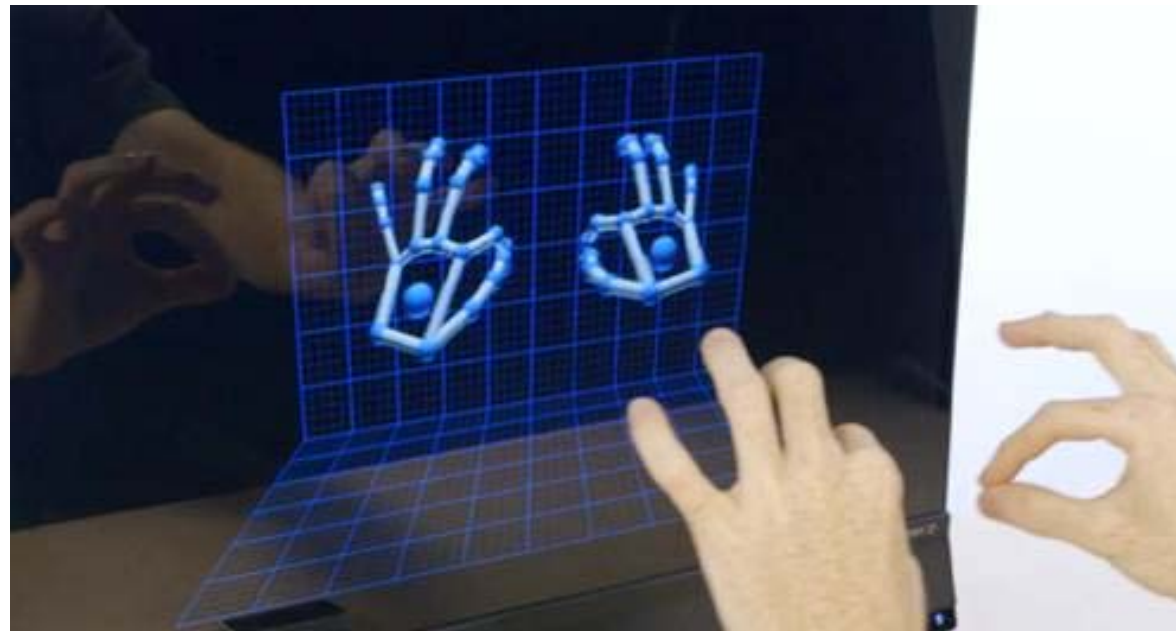
5,000
distinct

HAND gestures



Efficient Hand Tracking from “Depth” Images

- The problem:
 - Efficiently estimate a hand pose, with the skeleton or joints located



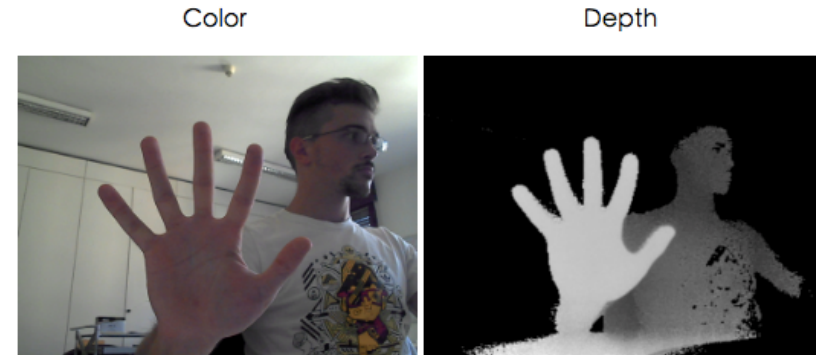
Gesture Recognition

- Challenges

- Hand's high degree of freedom, finger self-similarity, self-occlusion, etc.

- ChaLearn Competition (from 2011)

- <http://chalearnlap.cvc.uab.es/challenge/25/description/>



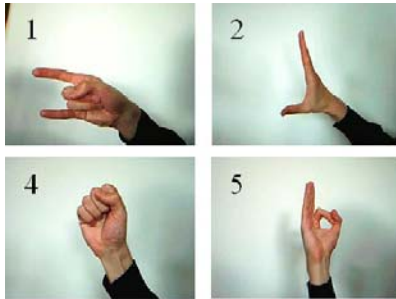
**Chalearn Coopetition on Action, Gesture, and Emotion Recognition: Large Scale
Multimodal Gesture Recognition and Real versus Fake expressed emotions
@ICCV17**



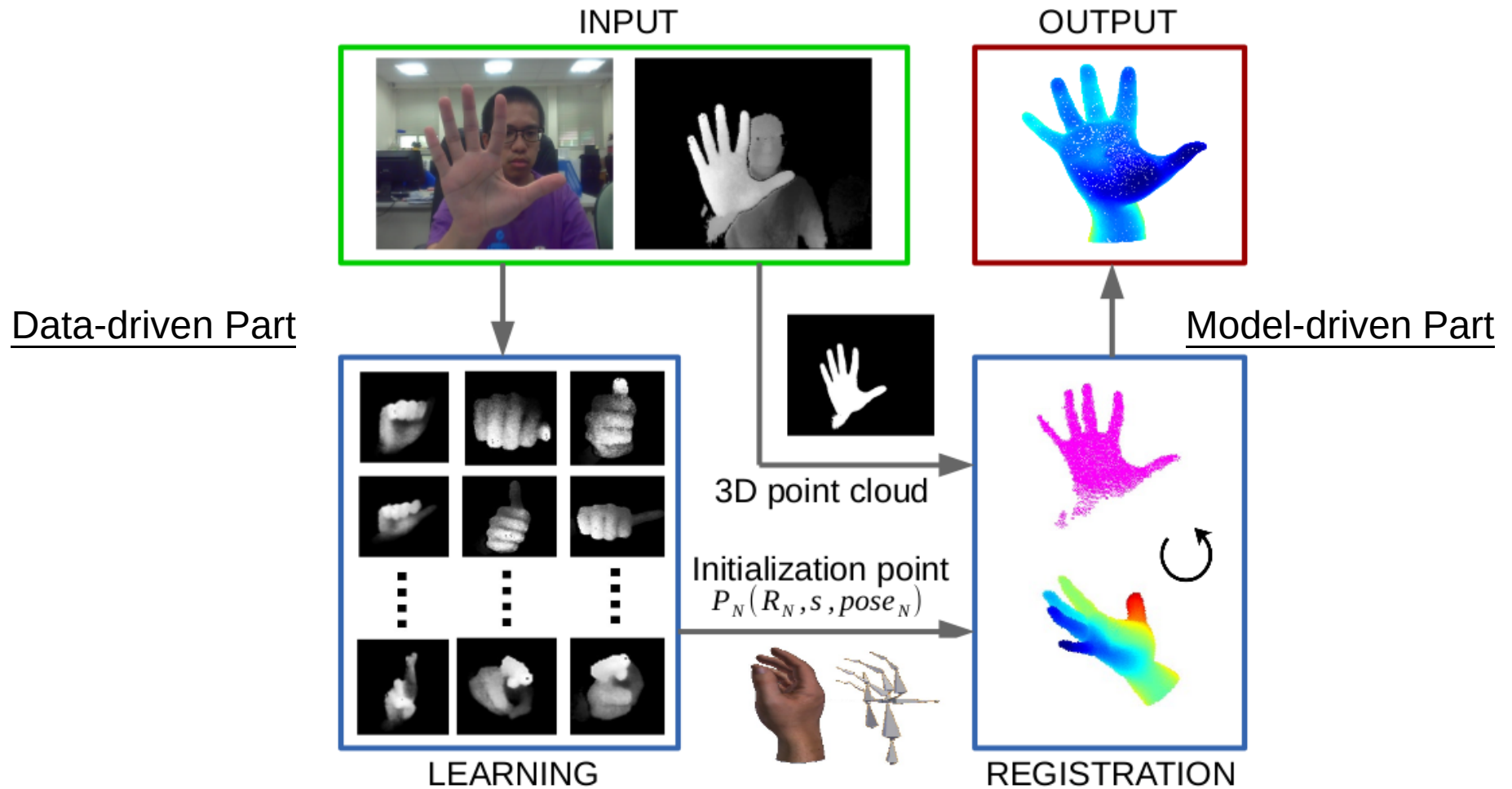
Existing Methods



- Model-driven approaches
 - To fit an explicit deformable hand model to depth images by minimizing a hand-crafted cost function, e.g., ICP (Iterative Closest Point)
- Data-driven approaches
 - To learn a mapping from depth image to hand pose from training data, e.g., CNN
- Hybrid approaches



Pipeline of the Method



Benchmark Development: Gesture

<http://mclab.citi.sinica.edu.tw/dataset/lared/lared.html>



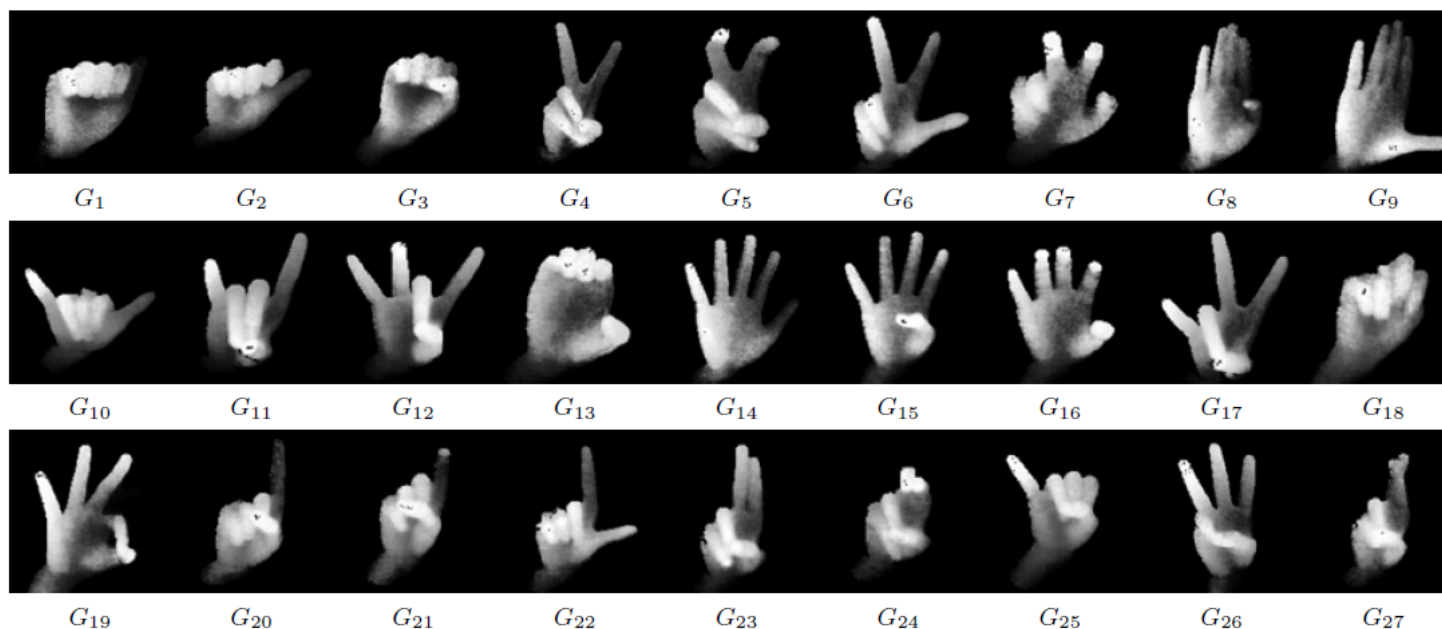
Large RGB-D Extensible Hand Gesture Dataset


National Taiwan University of
Science and Technology

Introduction	Gesture Classes/Rotations	Data Description	Download	Publications	Go to MCLab
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Introduction

We present the LaRED, a Large RGB-D Extensible Hand Gesture Dataset, recorded with an Intel's newly-developed short range depth camera. This dataset is unique and differs from the existing ones in several aspects. Firstly, the large volume of data recorded: 243,000 tuples with each is composed of a color image, a depth image, and a mask of the hand region. Secondly, the number of different classes provided: a total of 81 classes (27 gestures in 3 different rotations).



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Used by more than 60+ research teams worldwide...



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Heart Bus

Collaborators:  Taipei City Government
國立臺北藝術大學 Taipei National University of the Arts
 中國信託慈善基金會 CTBC Charity Foundation

An Experience Bus for Real-time Interactive Training of Rescuer Skills

- The dissemination of rescuer knowledge is crucial in public education
- The world's first experience bus for real-time interactive training of rescuer skills (CPR, AED, Heimlich), by leveraging the motion-sensing techniques



The bus is now circulating in Taipei City for public experience.



CPR



AED



Heimlich





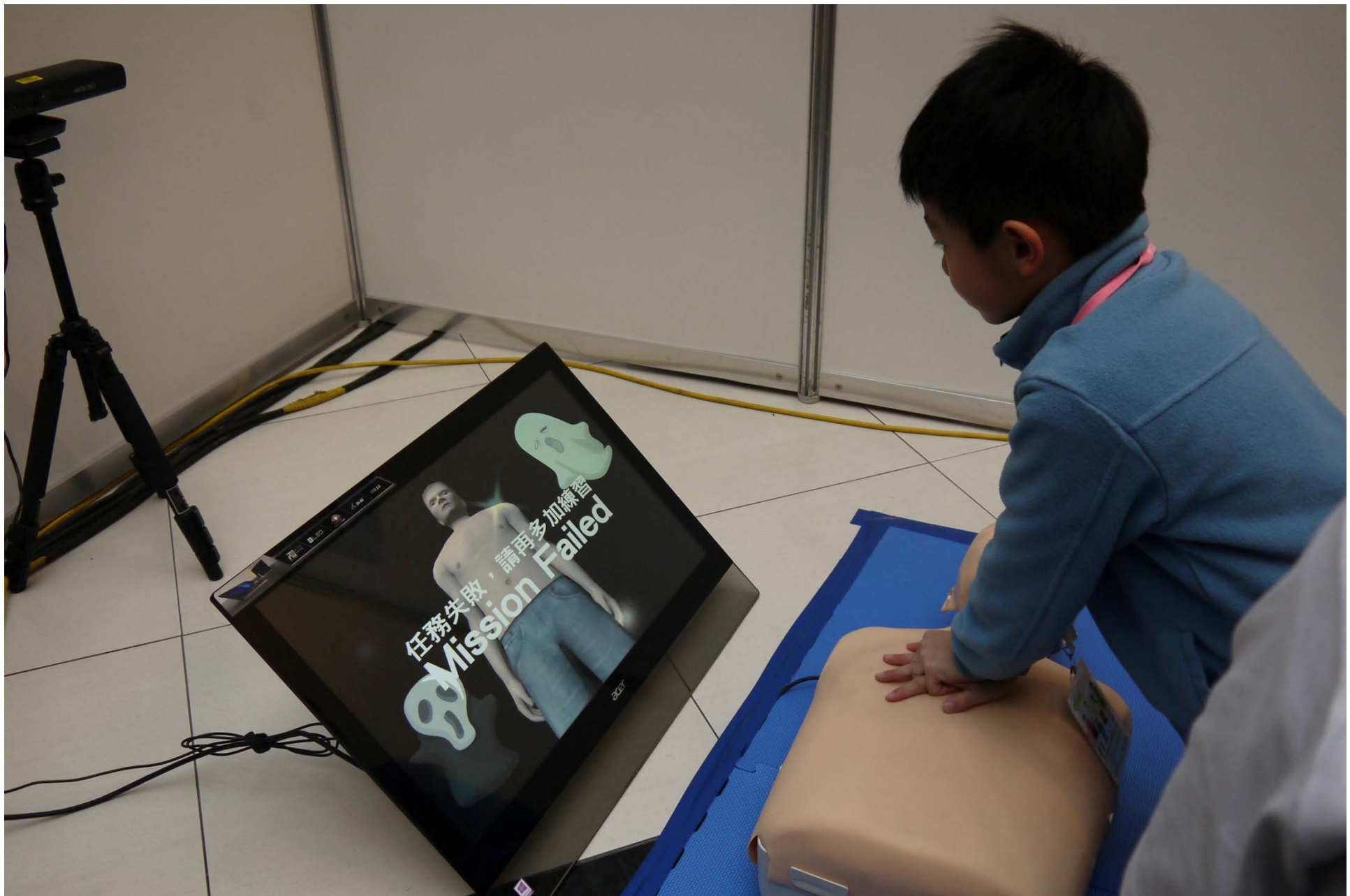
HeartBus opening by the Taipei City Mayor in the Taipei City Hall on 24 July, 2014





iCPR Demo



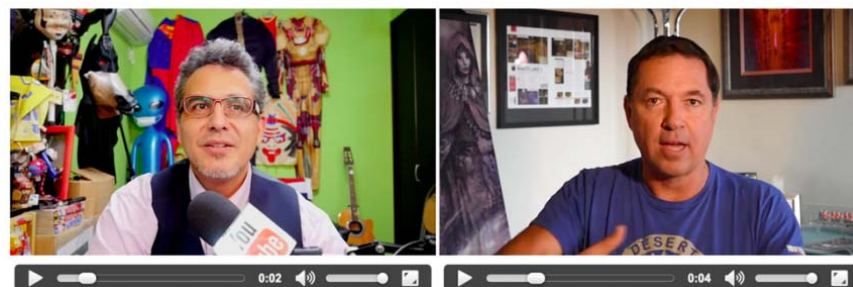


Visual Intelligence for Sociological Understanding

ChaLearn Workshop on Explainable Computer Vision Multimedia and **Job**
Candidate Screening Coopetition @CVPR17 and @IJCNN17

IJCNN - Anchorage, Alaska, May 2017

CVPR - Honolulu, Hawaii, July 2017



Please assign the following attributes to one of the videos:

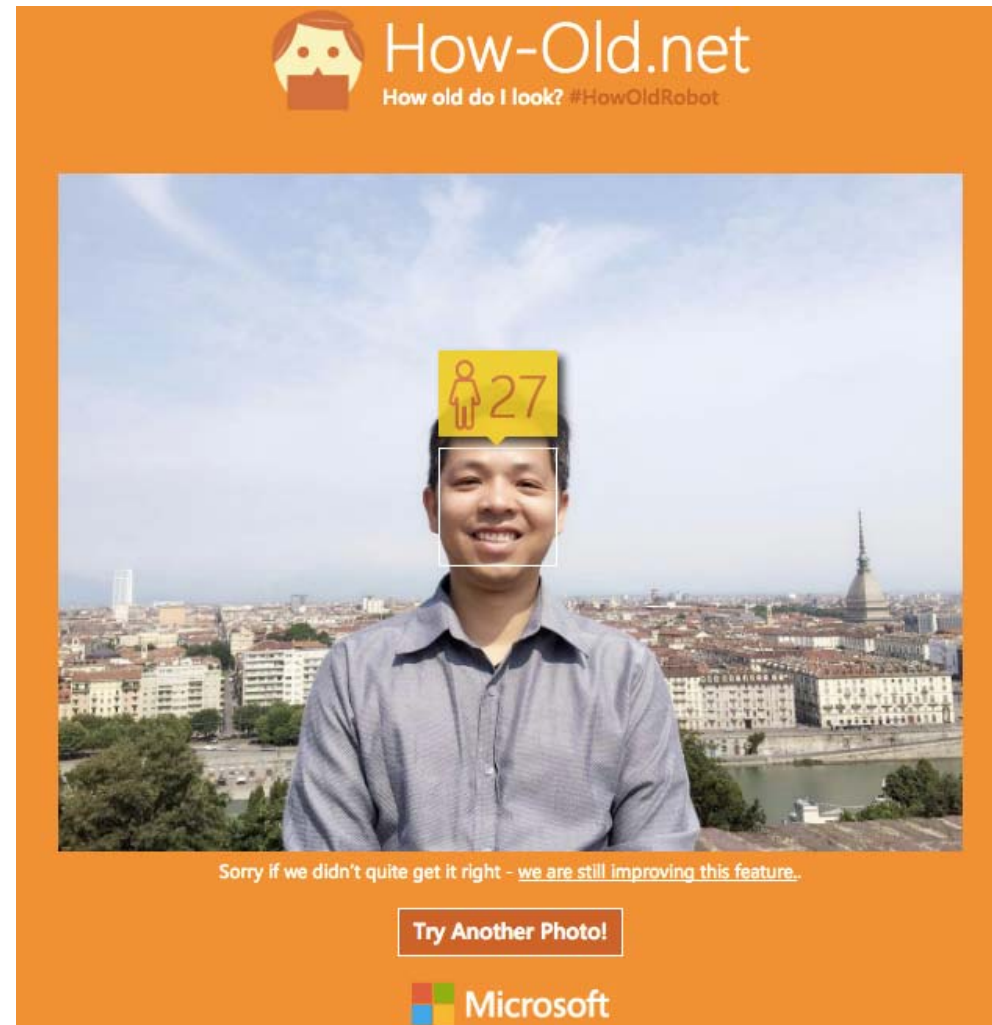
Friendly (vs. reserved)	Left	Don't know	Right
Authentic (vs. self-interested)	Left	Don't know	Right
Organized (vs. sloppy)	Left	Don't know	Right
Comfortable (vs. uneasy)	Left	Don't know	Right
Imaginative (vs. practical)	Left	Don't know	Right

Who would you rather invite for a job interview?

Left **Don't know** **Right**



Human Age Estimation



What's his age?

Matt Damon in 1997(27)



Difficulty to guess



Who's older?

Matt Damon in 1997(27)



>
=
<

Matt Damon in 2015(45)



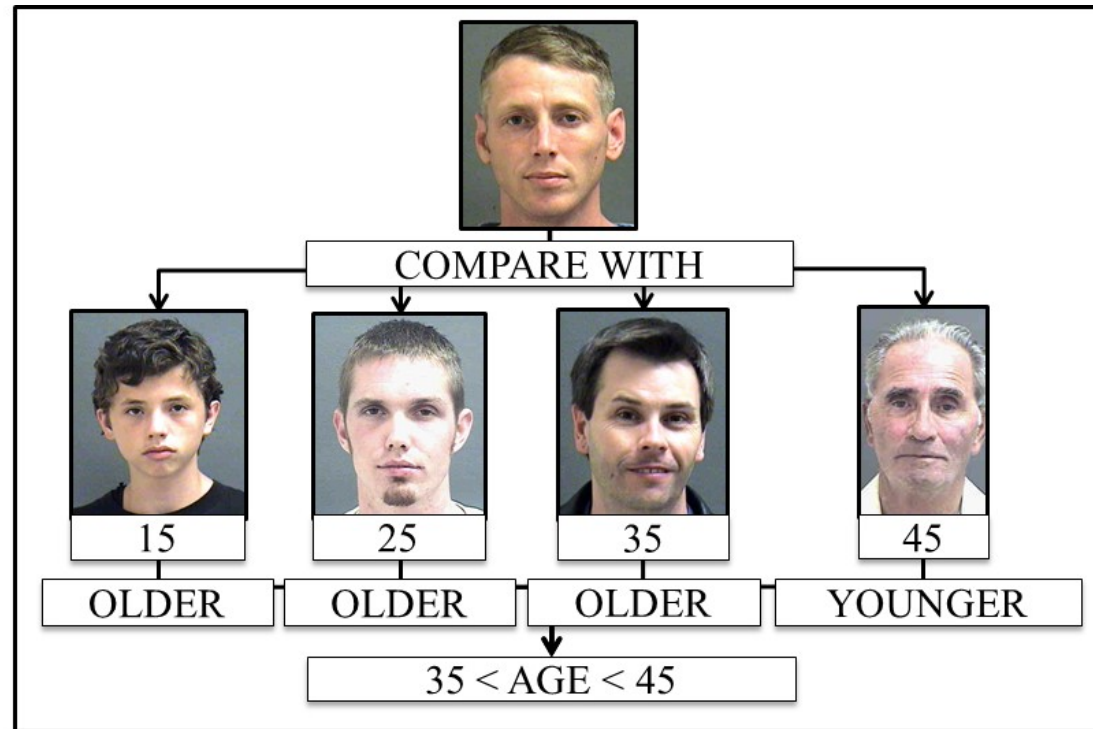
Easier to compare



The Proposed Comparative Paradigm



The Conventional Estimation



The Comparative Estimation

Clothing fashion is a reflection of the society of a period

- The global fashion apparel market today has surpassed 1 trillion US dollars since 2013, and accounts for nearly 2% of the world's Gross Domestic Product (GDP)

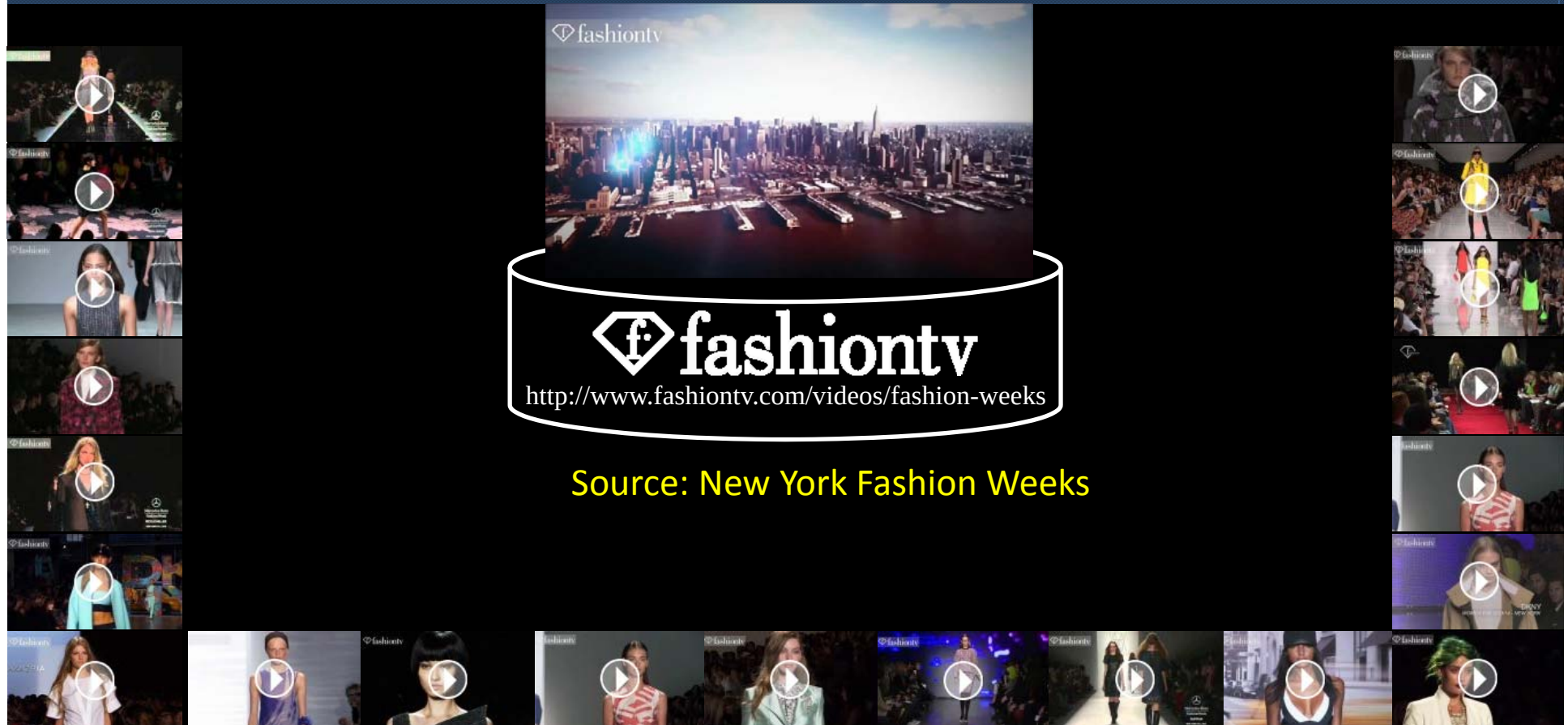


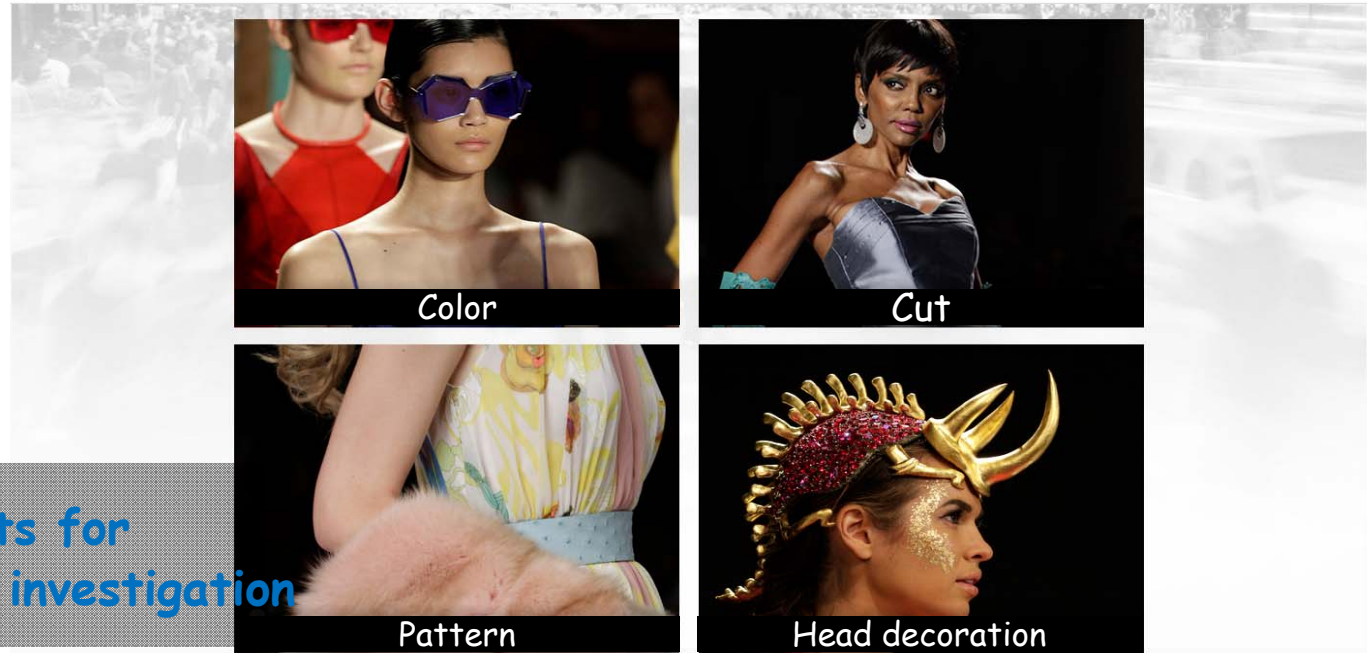
Trend Analysis for the Clothing Fashion

Our work received “Multimedia Grand Challenge Award” in 2014 ACM Multimedia Conference.



Construct a fashion show dataset





4

major elements for
fashion style investigation

2

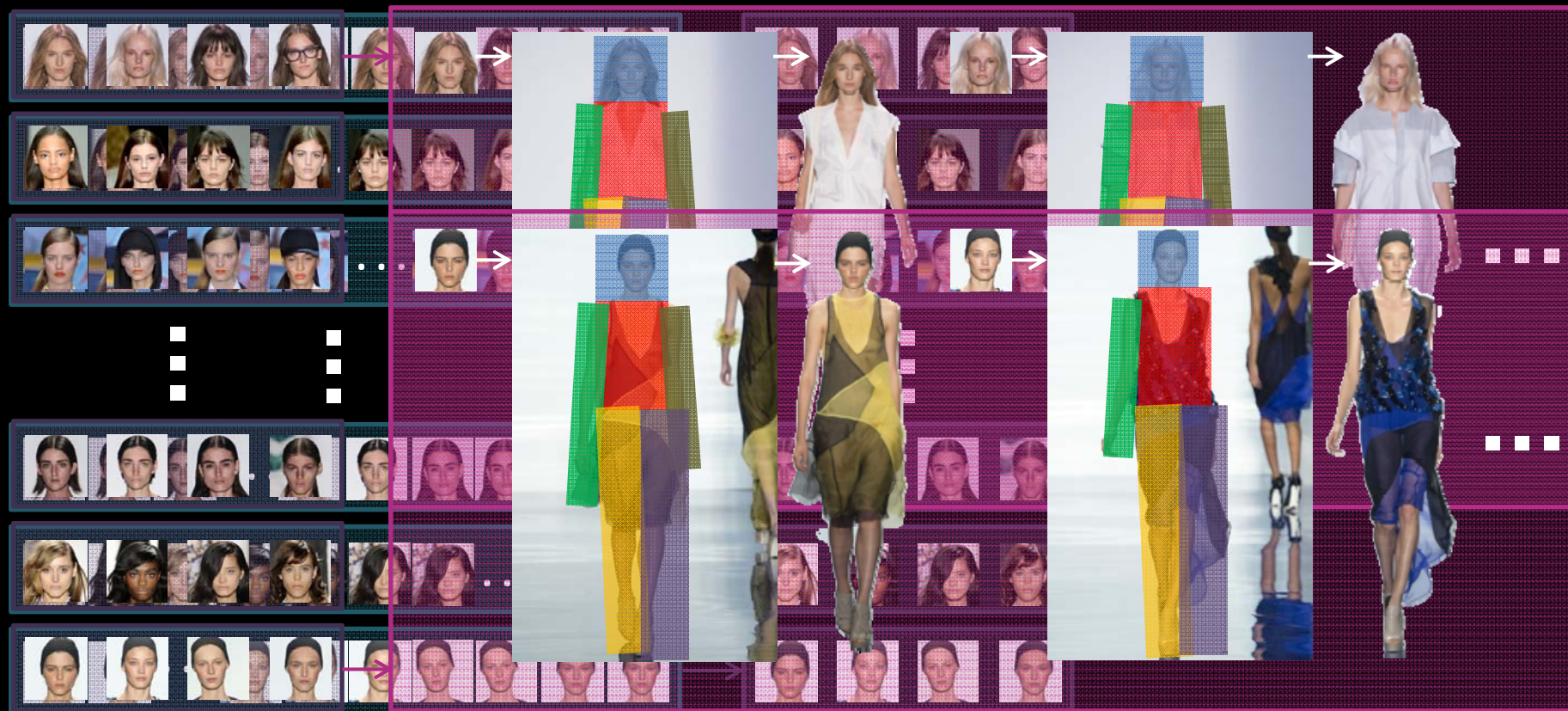
key factors for discovering fashion trends:

- ✓ coherence (frequently occur within a fashion week)
- ✓ uniqueness (occur much more often in a fashion week than in other fashion weeks)

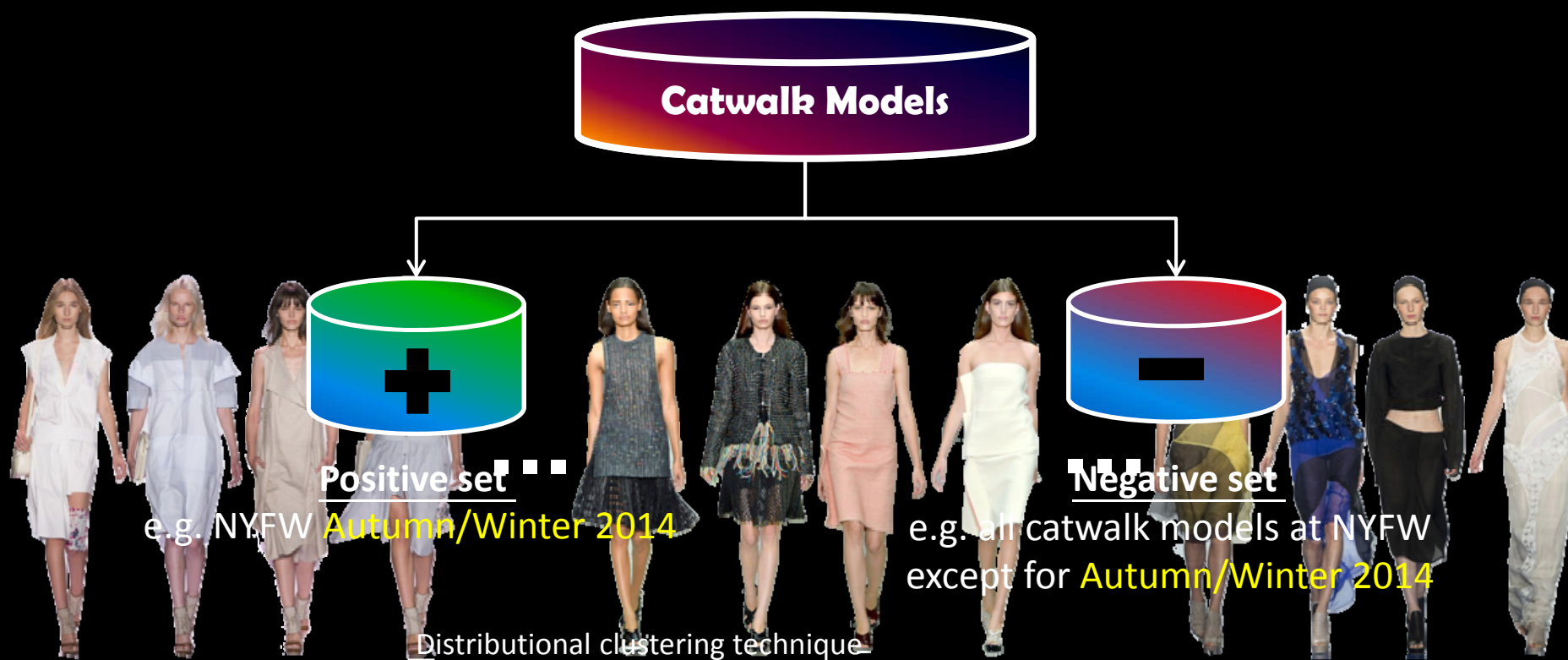
Detect the presence of distinct models in video frames



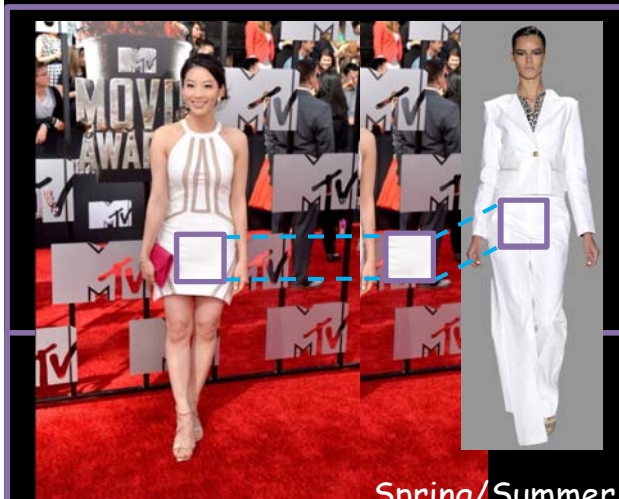
Extract identity location and the full body image



Divide the collected full-body catwalk images into two sets



W.-H. Cheng et al., "Learning and Recognition of On-Premise Signs (OPSs) from Weakly Labeled Street View Images," IEEE Tran. on Image Processing (TIP), 2014.



Spring/Summer
Query Image Image 2011

Color Analysis

✓ **Spring/Summer 2011**

Query Image

Color Analysis



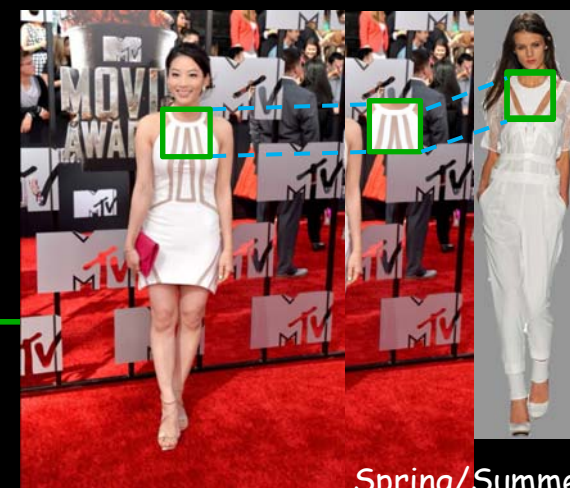
Spring/Summer
Query Image Image 2013

Texture Analysis

✓ **Spring/Summer 2013**

Query Image

Texture Analysis



Spring/Summer
Query Image Image 2013

Color + Texture Analysis

✓ **Spring/Summer 2013**

Query Image

Color + Texture Analysis

Can you tell where they are from?



Can you tell where they are from?



Fashion World Map: Understanding Cities Through Streetwear Fashion [MM17]



STREET & GENTLE LOOKB

93 Pins 80 Followers

Mens fashion

Streetwear

High fashion

Comment

Guy fashion

Style ide



Men's Black Suede Bomber Jacket, Black Tank, Grey Chinos, Tan Leopard Low



Yes or No? Follow @mensfashion_guide for more! By



LOOKBOOK

Explore

Feed

Blog

...

Star print in Kos

0 1 hour ago



HYPE 6

SAVE



Zaful Dress, Gucci Marmont



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GSFashion (Global Street Fashion) Dataset

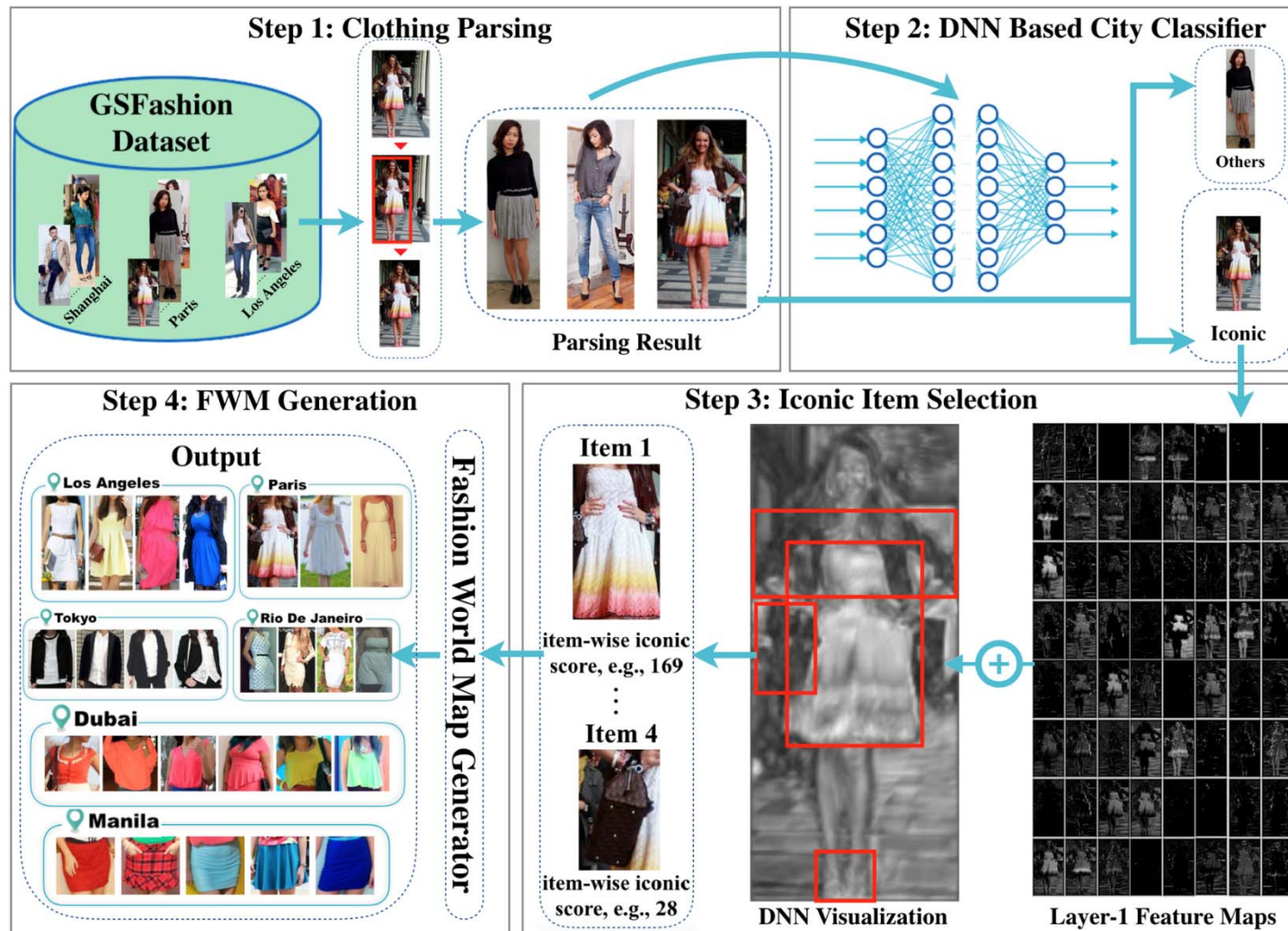
53 cities

North America: 8 | South America: 3 | Europe: 18 | Asia: 12 | Oceania: 2



The Analysis Pipeline

[MM17]



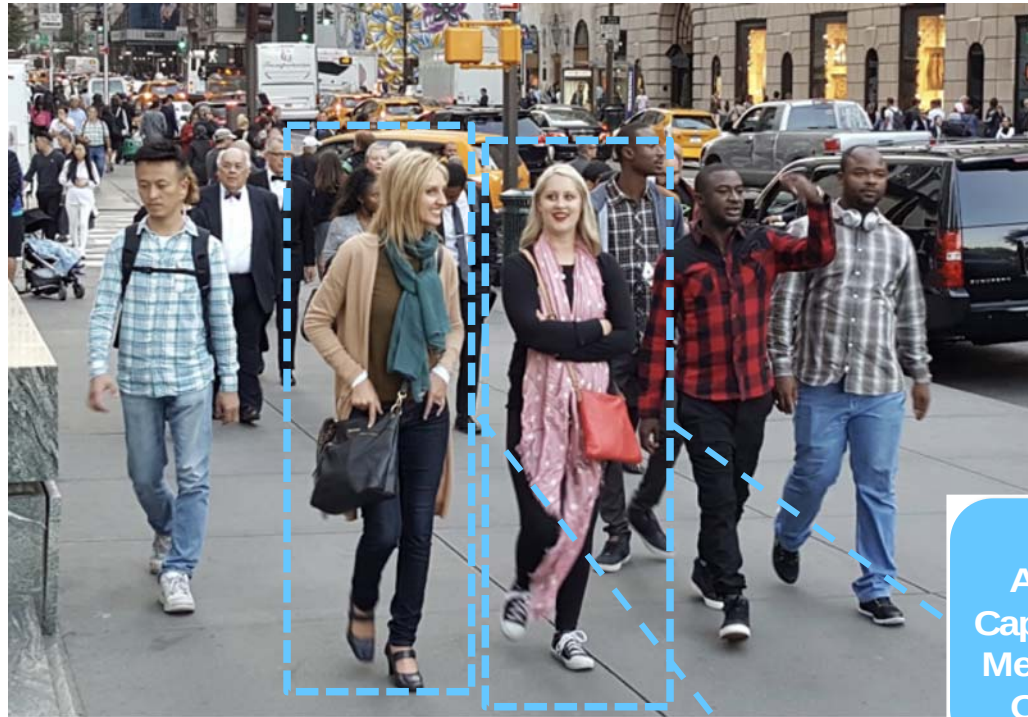
Fashion World Map

A visual summary of the city based street fashion



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“Judge people by their outfit appearance”



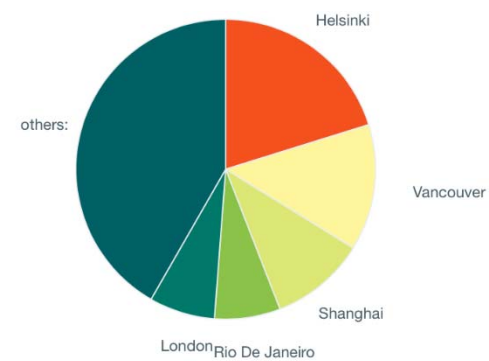
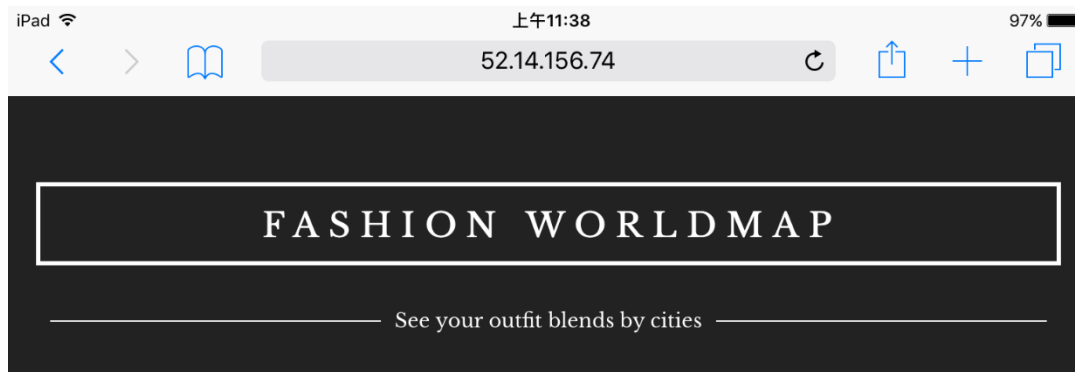
Paris: 38.6%
Athens: 17.2%
Cape Town: 10.8%
Mexico City: 9.6%
Others: 23.8%

Paris: 50.4%
New York: 10.2%
Mexico City: 7.4%
Sao Paulo: 5.1%
Others: 26.9%





fashionworldmap.citi.sinica.edu.tw



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PERFECT



HOME

CHALLENGE

AI Meets Beauty

Perfect Corp. Challenge 2018: Half Million Beauty Product Image Recognition



Main Organizers



Jau Huang

CyberLink Corp.



Johnny Tseng

Perfect Corp.



Wen-Huang
Cheng

Academia Sinica



Jia Jia

Tsinghua University



Si Liu

Chinese Academy of
Sciences



Jianlong Fu

Microsoft Research

**JAN
2017**

GLOBAL DIGITAL SNAPSHOT

KEY STATISTICAL INDICATORS FOR THE WORLD'S INTERNET, MOBILE, AND SOCIAL MEDIA USERS

TOTAL
POPULATION



7.476
BILLION

URBANISATION:
54%

INTERNET
USERS



3.773
BILLION

PENETRATION:
50%

ACTIVE SOCIAL
MEDIA USERS



2.789
BILLION

PENETRATION:
37%

MOBILE
SUBSCRIPTIONS



8.047
BILLION

vs. POPULATION:
108%

ACTIVE MOBILE
SOCIAL USERS



2.549
BILLION

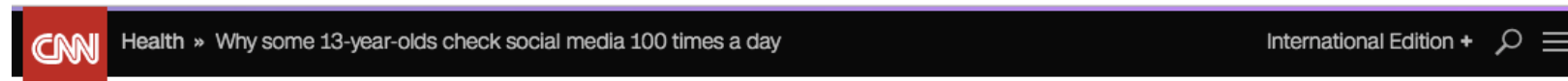
PENETRATION:
34%



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Sociology and Human Interaction

- With the huge number of people who are involved nowadays with social networks, it is very interesting to note how they are influenced by each other in many different ways.
 - e.g., **identity** in the age of social media



Why some 13-year-olds check social media 100 times a day

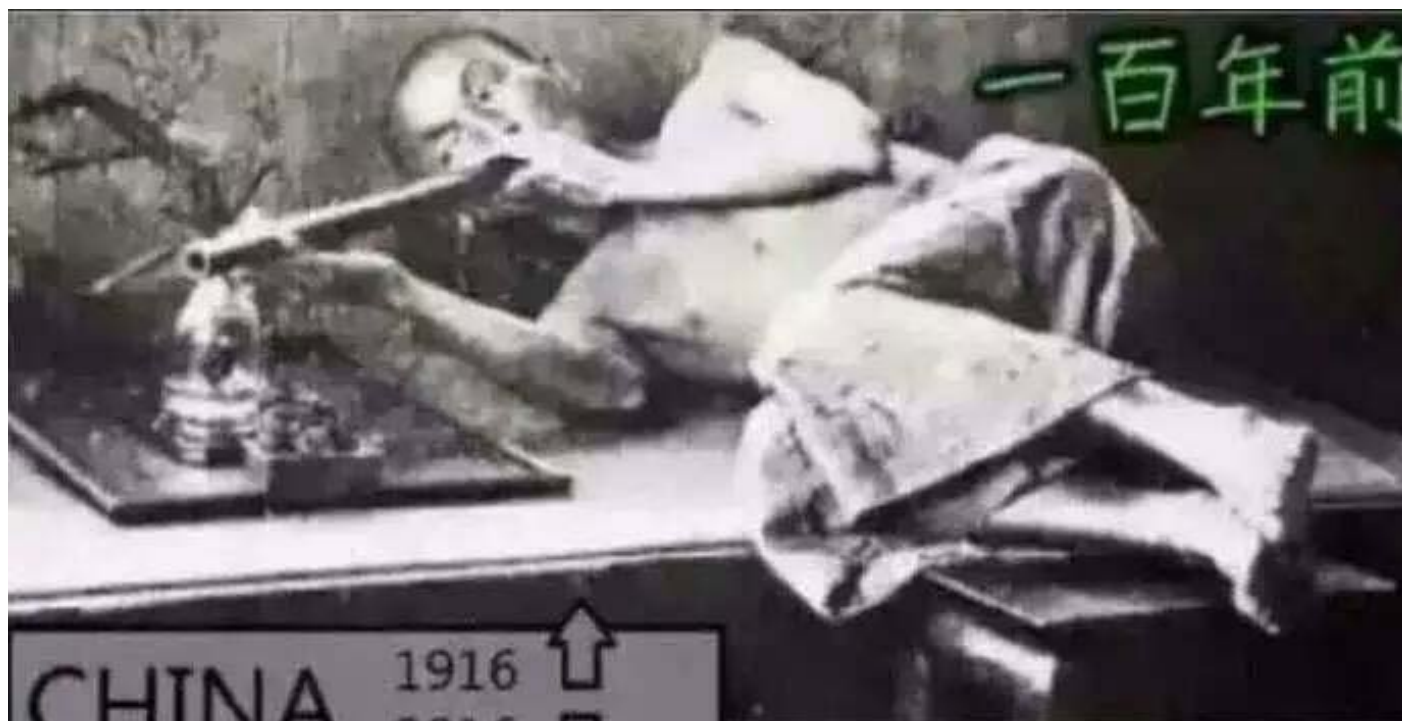
By **Chuck Hadad**, CNN
🕒 October 13, 2015

Why are teens so anxious about what's happening online? #Being13 found that it's largely due to a need to monitor their own popularity status, and defend themselves against those who challenge it.

- 61% of teens said they wanted to see if their online posts are getting likes and comments.
- 36% of teens said they wanted to see if their friends are doing things without them.
- 21% of teens said they wanted to make sure no one was saying mean things about them.

[Ref] <http://edition.cnn.com/2015/10/05/health/being-13-teens-social-media-study/index.html>

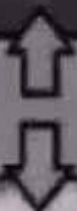




CHINA

1916

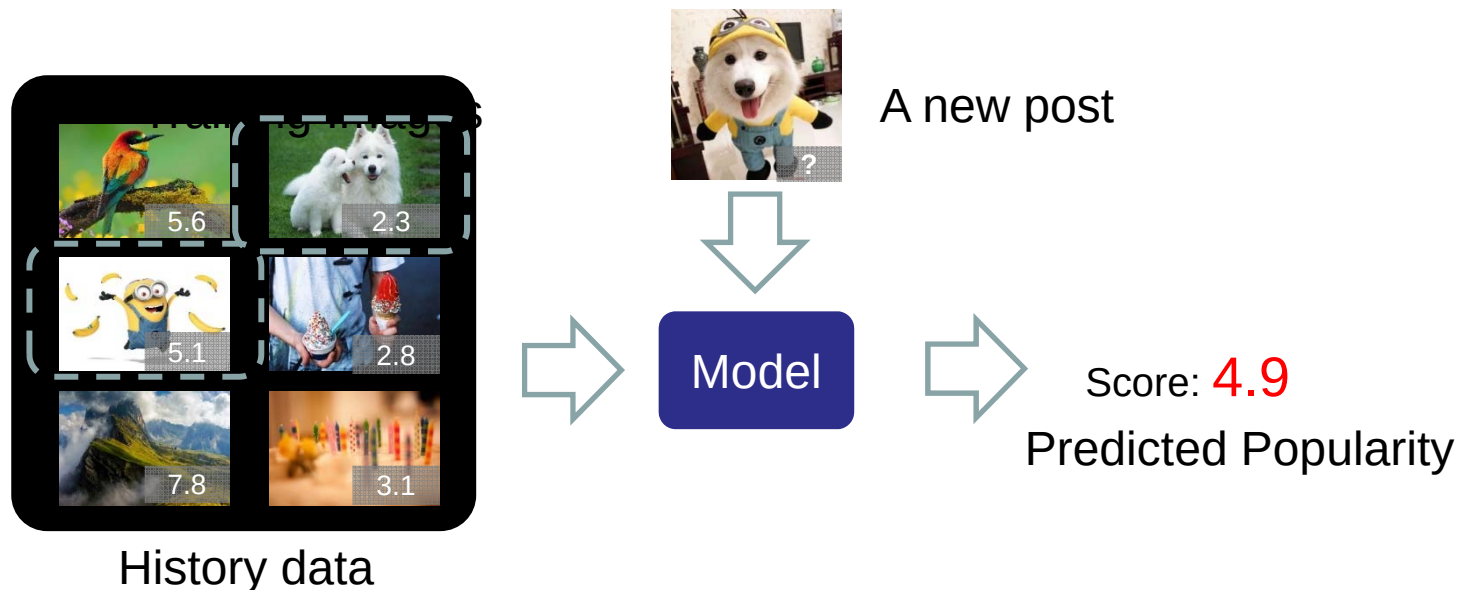
2016



@李开复

Social Popularity Prediction

- Predicting the **popularity score** of a new social media item (e.g., post) by combining **item content** (photo, text or video), **user cues**, and various features.



Why is it important?

- wide applications and high business value

Inside the Washington Post's popularity prediction experiment

A peek into the clickstream analysis and production pipeline for processing tens of millions of daily clicks, for thousands of articles.

By Shuguang Wang and Eui-Hong (Sam) Han, January 25, 2017

In the distributed age, news organizations shared more widely, potentially reaching a larger amount of time. At the *Washington Post*, we asked ourselves if it was possible to predict which stories will become popular. For the *Post* newsroom, this would be an invaluable tool, allowing editors to more

Why is it important?

- wide applications and high business value
 - e.g., predicting the “Stars of Tomorrow” (top popular models) within the fashion Industry using social media

Fashion Model Directory (FMD) profile page



KENDALL JENNER
AMERICAN

BORN NOVEMBER 3 1995
IN LOS ANGELES,
CALIFORNIA, USA

HAIR	EYES	HEIGHT	BUST	WAIST	HIPS	DRESS	SHOES
BROWN	BROWN	US/UK/UK 5'9" / 175CM	32 / 81	24 / 61	34 / 87	24 / 32-34	8.5 / 39 / 6

MOTHER AGENCY THE SOCIETY
NEW YORK

AGENCIES
ELITE MODEL MANAGEMENT - PARIS
ELITE MODEL MANAGEMENT - LONDON
WILHELMINA MODELS - PANAMA

MAGAZINE COVERS

CREDITED

2014

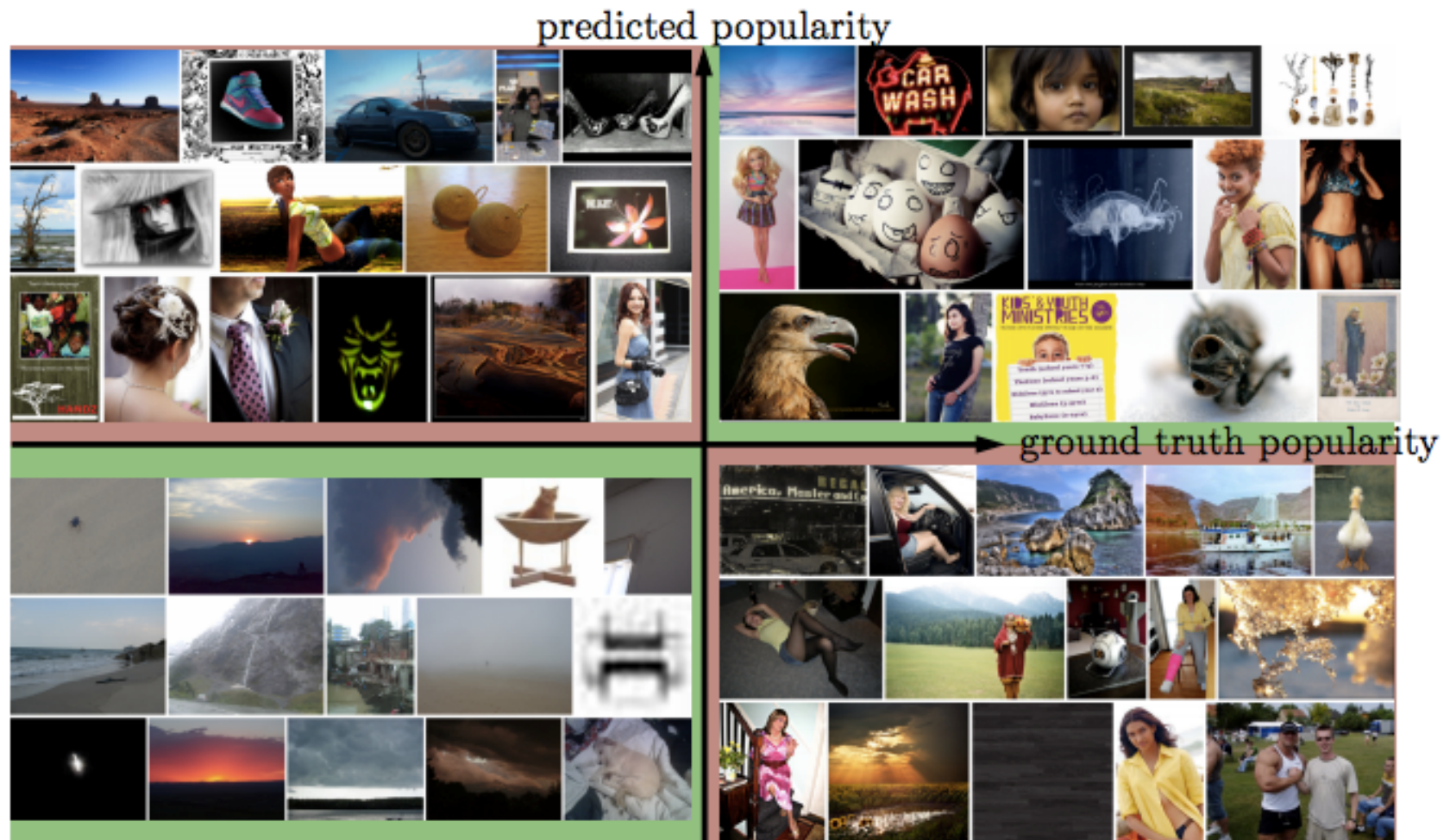
December Dazed
September LOVE, Teen Vogue USA, Teen Vogue USA
March Marie Claire Mexico

Can you tell who will be the “top”?



[Ref] “Style in the Age of Instagram: Predicting Success within the Fashion Industry using Social Media,” CSCW 2016.

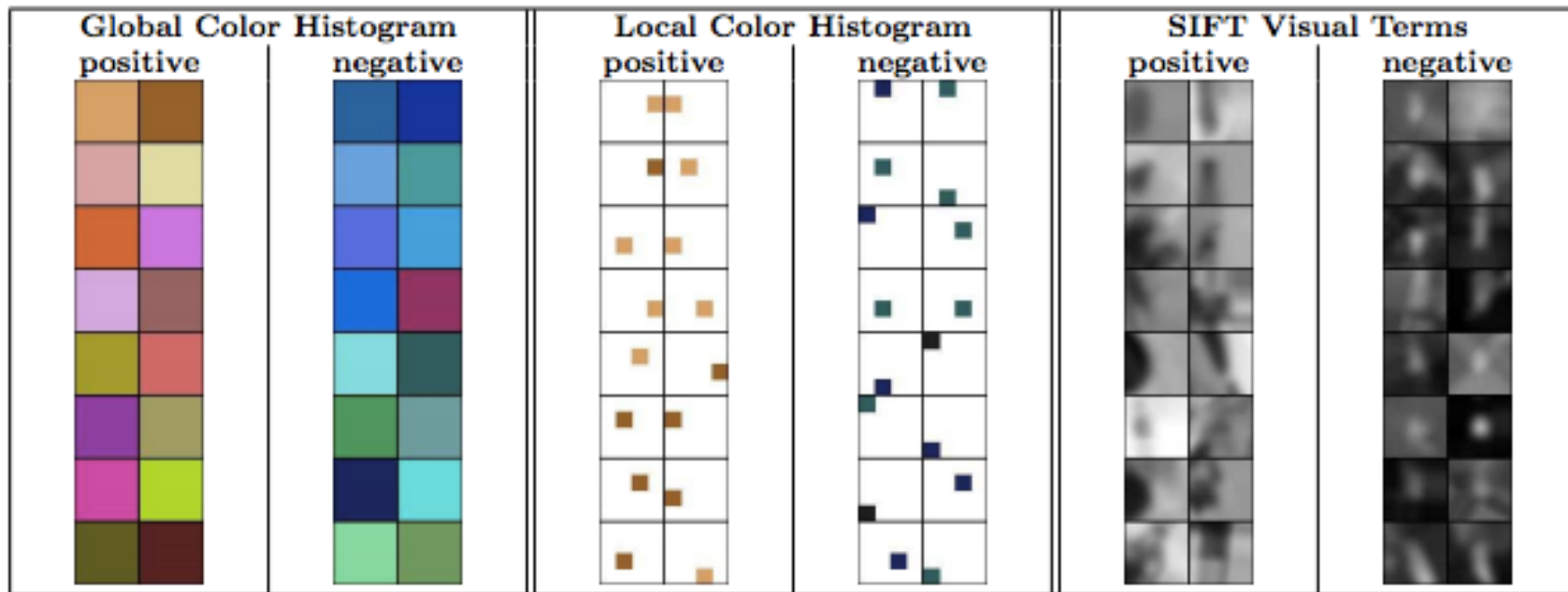
What Makes A Post Popular?



[Ref] “What Makes an Image Popular?” WWW, 2014.

What Makes A Post Popular?

- Features for prediction
 - Post content
 - ◆ e.g., visual sentiment features (color and texture)



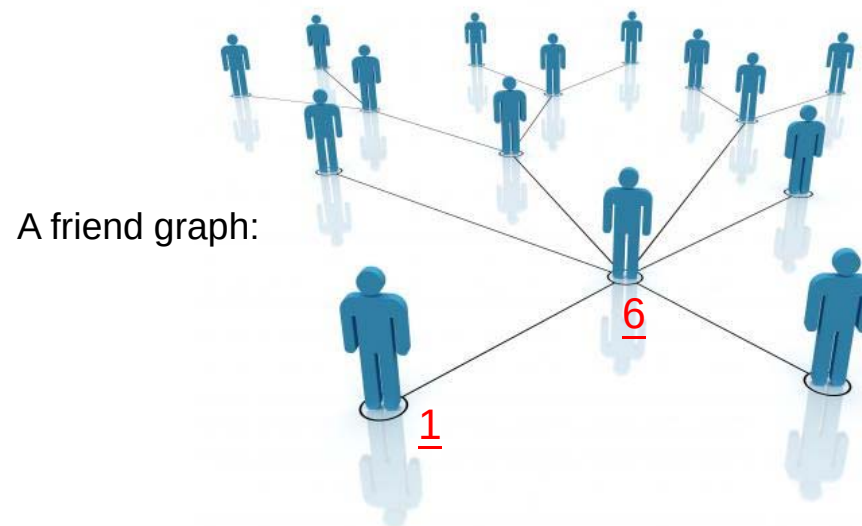
[Ref] "Analyzing and predicting sentiment of images on the social web," ACM Multimedia 2010.

What Makes A Post Popular?

- Features for prediction

- User cues

- ◆ e.g., **followers** (a user's follower count), **friends** (how many users a user follows), **statuses** (a user's current total post count), **user time** (a user's account creation time), etc.



Time Matters:

Five Popular Articles @ Washington Post

Articles	# clicks @ 30mins	# clicks @ 24hours
9/11 Flag	6,245	67,028
Trump Policy	2,015	128,217
North Carolina	1,952	11,406
Hillary & Trump	1,733	310,702
Gary Johnson	1,318	196,798

Gary Johnson

- Summary: It would be easy to describe Gary Johnson's appearance on MSNBC's 'Morning Joe'.
- 30 mins: 1,318 clicks



Hillary & Trump

- Summary: There was a very weird thread worth picking out of Donald Trump's comments during NBC's 'Commander-in-Chief Forum'.
- 30 mins: 1,733 clicks



9/11 Flag

- Summary: Shortly after three New York firefighters raised an American flag above the ruins left by the Sept. 11 attacks — a moment captured in an iconic photo seen around the world — the flag disappeared.
- 30 mins: 6,245 clicks



Trump Policy

- Summary: The Trump campaign built a large policy shop in Washington that has now largely melted away because of neglect, mismanagement and promises of pay that were never honored.
- 30 mins: 2,015 clicks



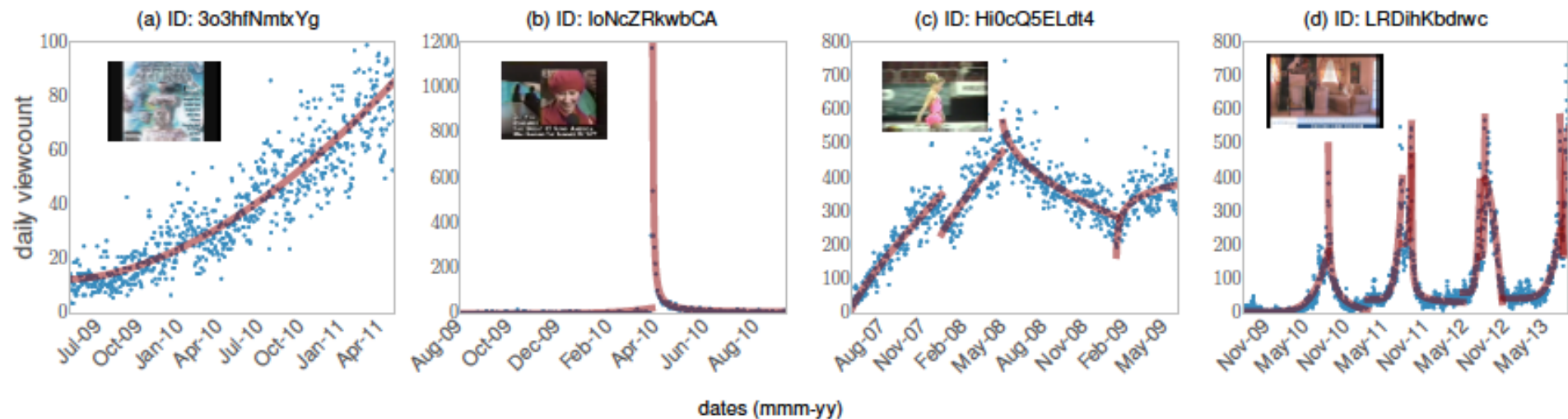
North Carolina

- Summary: Two bodies and a burning house on the outskirts of a small North Carolina town were the result of a grisly altercation between a Marine, his estranged wife and her boyfriend late last month.
- 30 mins: 1,952 clicks



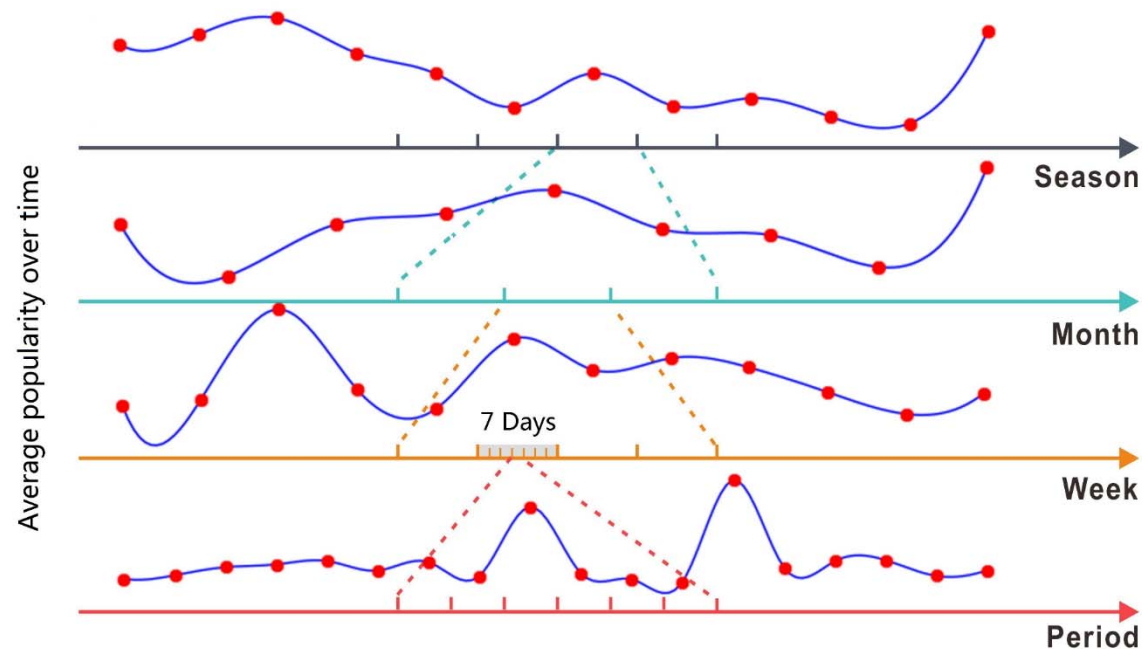
Temporal Modeling for Popularity

- To incorporate the temporal evolving structures in popularity prediction



Challenge 1: Temporal Evolving

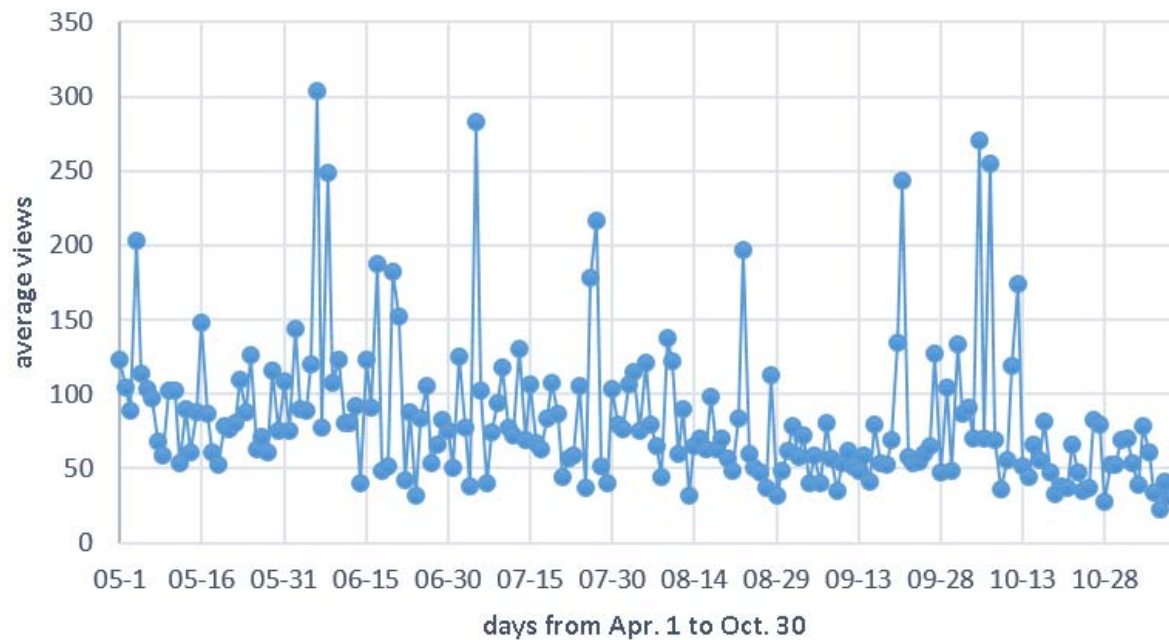
- The popularity evolving at multi-granularities with different patterns



Multi-granularities Characteristics of Popularity Dynamics

Challenge 2: Data Noise

- Popularity patterns are covered in very noisy behavior data or information

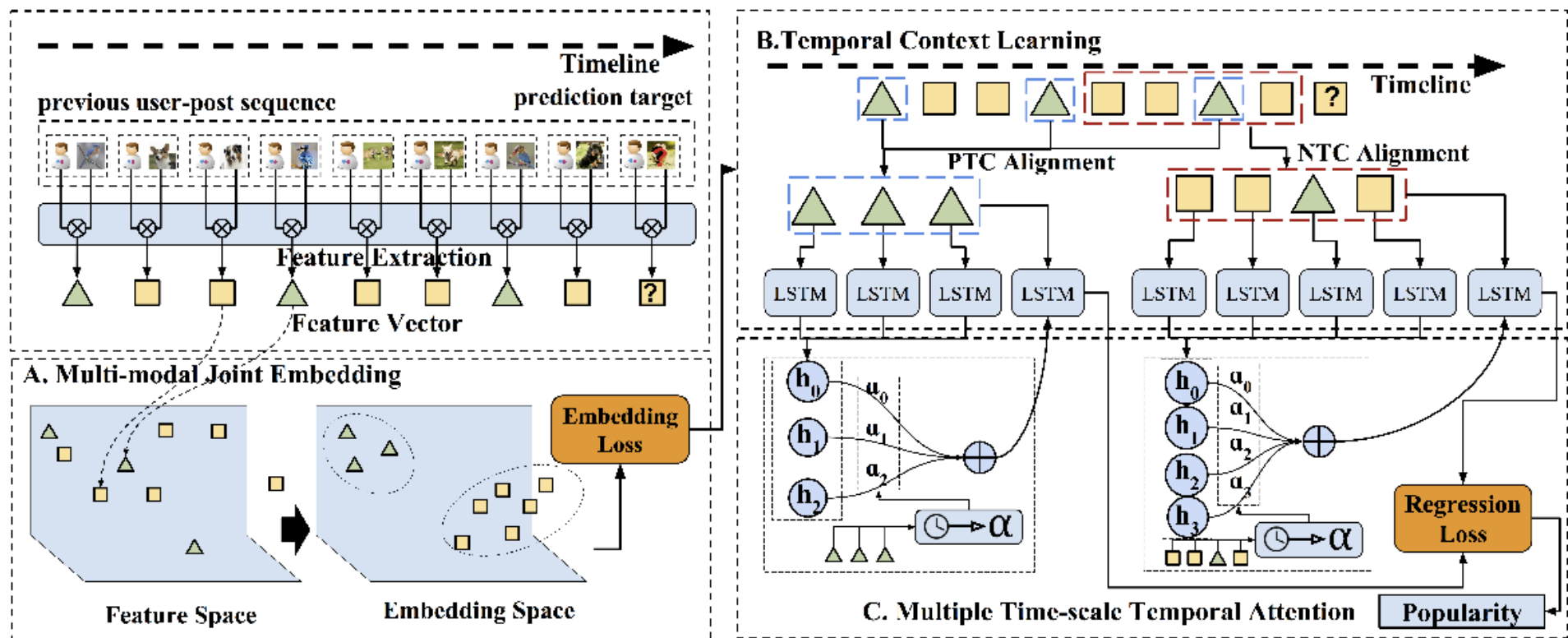


Popularity distribution on time series

Deep Learning for Modeling the Prediction

[IJCAI2017]

- We address the problem as a sequential prediction task, where the input is a user-photo sequence (with time order) while the output is the popularity of a “future” photo (a photo before its publication on social media)



Experiments

- Data Sets

- Over 1.8M photos
- Over 70K users
- Views, User profile, Photo stream
- Metadata, Images, Annotations



- Setting

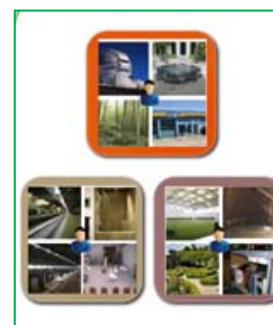
Photo-mix Dataset (PmD)

Users 70K
Images 1200K



User-specific Dataset (UsD)

Users 400
Images 600K



LFM

Method	40K UsD	750K UsD	30K UmD	540K UmD
Average Views	0.0574	0.0854	0.1342	0.3202
Bipartite Graph	0.2286	0.2686	0.2345	0.6895
SVR	0.1958	0.2522	0.2316	0.7432
BasicMF	0.1043	0.2425	0.2783	0.4822
PreferenceMF	0.1835	0.2816	0.3172	0.7532
MTD	0.2462	0.3491	0.3425	0.7977

MT

Method	400K PmD	800K PmD	300K UaD	600K UaD
AV	0.0419	0.0785	0.1043	0.2022
LR	0.2805	0.3026	0.5582	0.6359
SVR	0.2627	0.2725	0.4415	0.6273
BG	0.2385	0.2573	0.2851	0.5465
TMF	0.2186	0.2839	0.4352	0.5795
MT (ours)	0.2916	0.3264	0.5972	0.6574

DTCN

Dataset	100K	200K	400K
Metric	SRC		
CNN-AlexNet	0.2450	0.2435	0.1423
CNN-VGG	0.2281	0.2445	0.1064
SVR	0.2647	0.2367	0.2289
SVR(T)	0.2741	0.2984	0.2643
MLP	0.4135	0.5282	0.5559
MLP(T)	0.4436	0.5068	0.5084
LSTM	0.4629	0.5837	0.5885
CLSTM	0.4966	0.5730	0.6072
DTCN	0.5990	0.6175	0.6692

ACM Multimedia 2017

<http://www.acmmm.org/2017/>



Grand Challenge

Social Media Prediction (SMP): Predicting the “Stars of Tomorrow” on Social Media

<https://social-media-prediction.github.io/MM17PredictionChallenge/>

Organizers



Wen-Huang Cheng

Academia Sinica



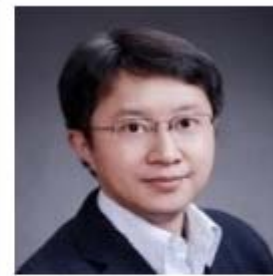
Bo Wu

Chinese Academy of
Sciences



Yongdong Zhang

Chinese Academy of
Sciences



Tao Mei

Microsoft Research
Asia

Sponsors



ACM Multimedia 2018

<http://www.acmmm.org/2018/>



Grand Challenge

Social Media Headline Prediction (SMHP):

Getting a deeper understanding of the temporal evolution of social media headlines

<https://social-media-prediction.github.io/PredictionChallenge/>

Organizers



Wen-Huang Cheng

Academia Sinica



Bo Wu

JD AI Research



Tao Mei

JD AI Research

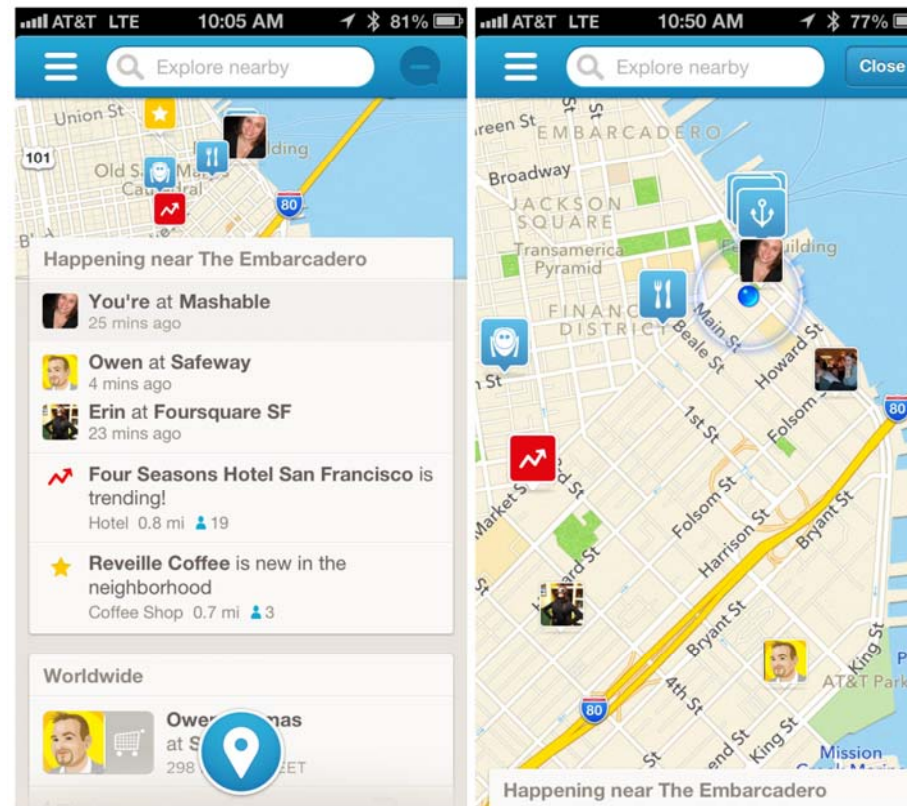
Sponsors



Foursquare Dataset

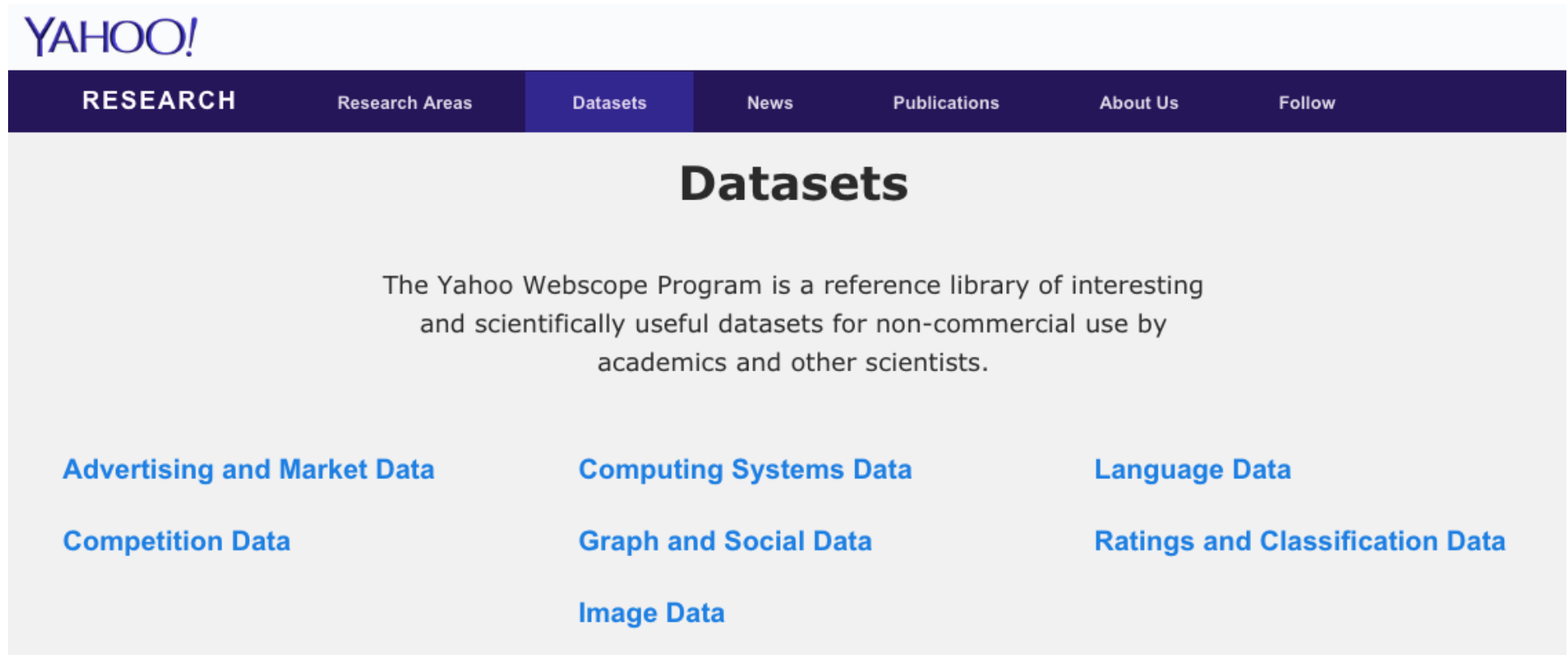
<https://sites.google.com/site/yangdingqi/home/foursquare-dataset>

- Individual “check-ins” data of the more than 10 million users on Foursquare



Yahoo! Dataset

<http://webscope.sandbox.yahoo.com/>



The screenshot shows the Yahoo! Research Datasets page. At the top is the Yahoo! logo. Below it is a dark blue navigation bar with the word "RESEARCH" in white, followed by links for "Research Areas", "Datasets" (which is highlighted), "News", "Publications", "About Us", and "Follow". The main content area has a light gray background. The word "Datasets" is centered in a large, bold, black font. Below it, a paragraph states: "The Yahoo Webscope Program is a reference library of interesting and scientifically useful datasets for non-commercial use by academics and other scientists." Further down, there are seven links arranged in three columns: "Advertising and Market Data", "Computing Systems Data", "Language Data", "Competition Data", "Graph and Social Data", "Ratings and Classification Data", and "Image Data".

YAHOO!

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Datasets

The Yahoo Webscope Program is a reference library of interesting and scientifically useful datasets for non-commercial use by academics and other scientists.

[Advertising and Market Data](#) [Computing Systems Data](#) [Language Data](#)

[Competition Data](#) [Graph and Social Data](#) [Ratings and Classification Data](#)

[Image Data](#)



YFCC100M

- This dataset contains 100 million media objects and explain the rationale behind its creation. This list is compiled from data available on Yahoo! Flickr.



(a) IMG_9793: Streetcar (Toronto Transit) by Andy Nystrom © ⓘ ⓘ ⓘ <https://flic.kr/p/jciMdz>.



(b) Celebrating our 6th wedding anniversary in Villa Mary by Rita & Tomek © ⓘ ⓘ ⓘ ⓘ <https://flic.kr/p/fCXEJi>.

Two photos of real world scenes from photographers in the YFCC100M dataset.

YFCC100M

- Global coverage of a sample of one million photos from the YFCC100M dataset.



Yelp Dataset Challenge

https://www.yelp.com/dataset_challenge



Yelp Dataset Challenge

Round 7 Of The Yelp Dataset Challenge: Now With Photos!

We've had 6 rounds, over \$40,000 in cash prizes awarded, [hundreds of academic papers written](#), and we are excited to see round 7.

The Challenge Dataset:

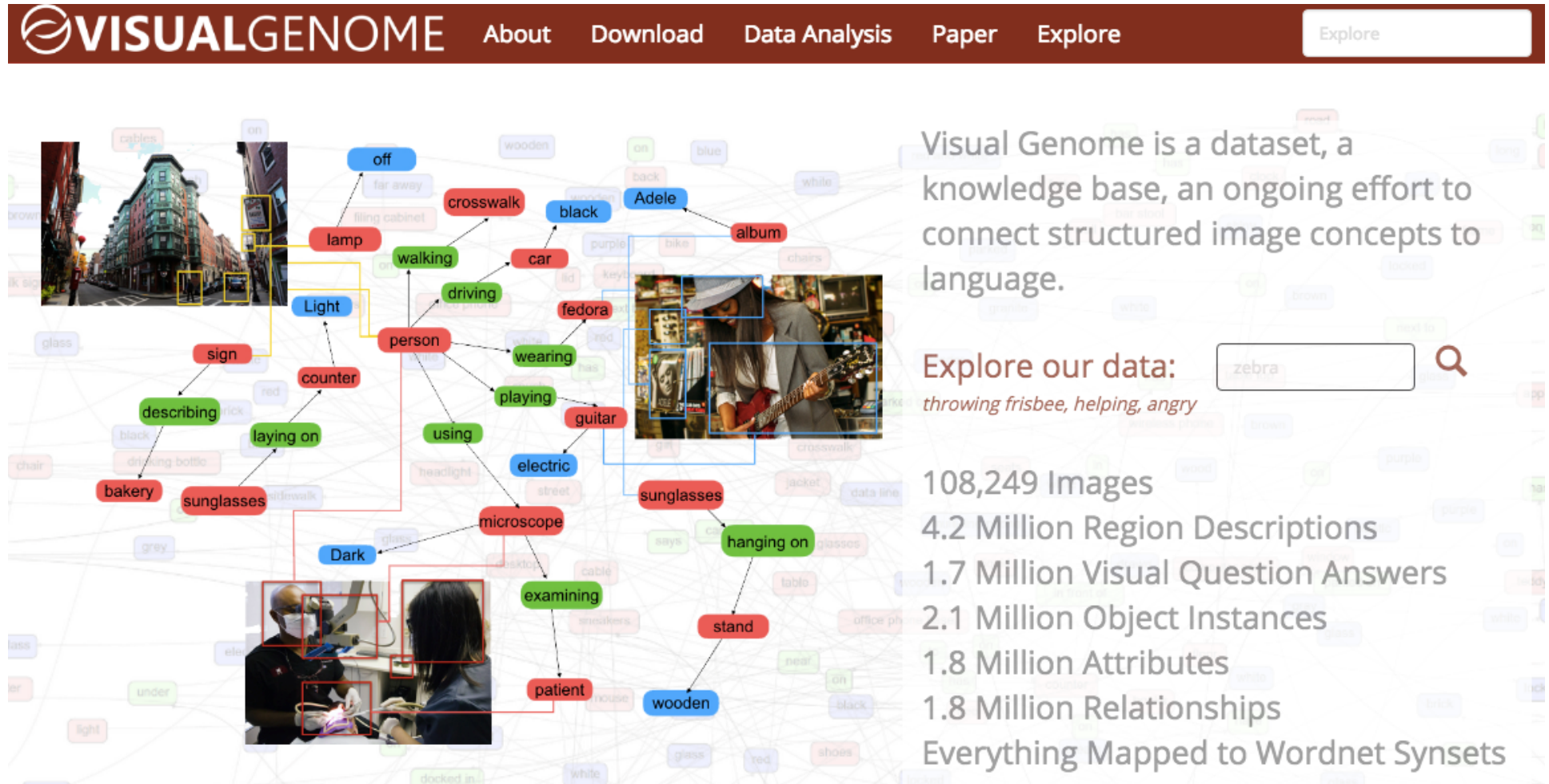
- **2.2M** reviews and **591K** tips by **552K** users for **77K** businesses
- **566K** business attributes, e.g., hours, parking availability, ambience.
- Social network of **552K** users for a total of **3.5M** social edges.
- Aggregated check-ins over time for each of the **77K** businesses
- **200,000** pictures from the included businesses

Cities:

- U.K.: Edinburgh
- Germany: Karlsruhe
- Canada: Montreal and Waterloo
- U.S.: Pittsburgh, Charlotte, Urbana-Champaign, Phoenix, Las Vegas, Madison

Visual Genome Dataset

<https://visualgenome.org/>



Instagram Dataset

<http://www.emilio.ferrara.name/datasets/>

Emilio Ferrara, Ph.D.

Research Asst. Prof. @ Univ. of Southern California

Source: Public media and user information from Instagram.com (through Instagram API).

Crawling period: Jan 20 – Feb 17, 2014.

Description: The media dataset contains records of the form: the anonymized media ID, the anonymized ID of the user who created the media, the timestamp of media creation, the set of tags assigned to the media, the number of likes and the number of comments it received. The anonymized user network contains asymmetric relations (A follows B); each edge is associated with #likes (by A to media created by B), #comments and the list of comments' timestamps.

Size

Media dataset: 1.7M media associated to 2K users, with 9M tags, 1200M likes, and 41M comments.

User network: about 45K vertices and 678K edges.

ICWSM-16 Dataset

International Conference on Web and Social Media

- <http://www.icwsm.org/2016/datasets/datasets/>

- **iFeel 2.0: A Multilingual Benchmarking System for Sentence-Level Sentiment Analysis**
Matheus Araújo, João P. Diniz, Lucas Bastos, Elias Soares, Manoel Júnior, Miller Ferreira, Filipe Ribeiro, Fabrício Benevenuto

- **Analyzing Personality through Social Media Profile Picture Choice**
Legi Liu, Daniel Preotiuc-Pietro, Zahra Riahi Samani, Mohsen E. Moghaddam, Lyle Ungar

- **The News Cycle's Influence on Social Media Activity**
Andrew Yates, Jonah Joselow, Nazli Goharian

- **Social Media Participation in an Activist Movement for Racial Equality**
Munmun De Choudhury, Shagun Jhaver, Benjamin Sugar, Ingmar Weber

- **EigenTransitions with Hypothesis Testing: The Anatomy of Urban Mobility**
Ke Zhang, Yu-Ru Lin, Konstantinos Pelechrinis

- **Discovering Response-Eliciting Factors in Social Question-Answering: A Reddit Inspired Study**
Danish, Yogesh Dahiya, Partha Talukdar

- **Distinguishing between Topical and Non-Topical Information Diffusion Mechanisms in Social Media**
Przemysław A. Grabowicz, Niloy Ganguly, Krishna P. Gummadi

- **The Road to Popularity: The Dilution of Growing Audience on Twitter**
Przemysław A. Grabowicz, Mahmoudreza Babaei, Juhi Kulshrestha, Ingmar Weber

- **Expertise in Social Networks: How Do Experts Differ from Other Users?**
Benjamin D. Horne, Dorit Nevo, Jesse Freitas, Heng Ji, Sibel Adali

- **On the Behaviour of Deviant Communities in Online Social Networks**
Mauro Coletto, Luca Maria Aiello, Claudio Lucchese, Fabrizio Silvestri

- **Your Age Is No Secret: Inferring Microbloggers' Ages via Content and Interaction Analysis**
Jinxue Zhang, Xia Hu, Yanchao Zhang, Huan Liu

- **Twitter's Glass Ceiling: The Effect of Perceived Gender on Online Visibility**
Shirin Nilizadeh, Anne Groggel, Peter Lista, Srijita Das, Yong-Yeol Ahn, Apu Kapadia, Fabio Rojas

- **Fusing Audio, Textual and Visual Features for Sentiment Analysis of News Videos**
Moisés H. R. Pereira, Flávio L. C. Pádua, Adriano C. M. Pereira, Fabrício Benevenuto, Daniel H. Dalip

- **Science, AskScience, and BadScience: On the Coexistence of Highly Related Communities**
Jack Hessel, Chenhao Tan, Lillian Lee

- **Comparing Overall and Targeted Sentiments in Social Media during Crises**
Saúl Vargas, Richard McCreadie, Craig Macdonald, Iadh Ounis



MediaEval Competition



MediaEval Benchmark

MediaEval Benchmarking Initiative for Multimedia Evaluation

The "multi" in multimedia: speech, audio, visual content, tags, users, context

- MediaEval is a multimedia benchmark evaluation that offers tasks promoting research and innovation in areas related to human and social aspects of multimedia.



IEEE International Conference on Visual Communications and Image Processing

IEEE VCIP 2018

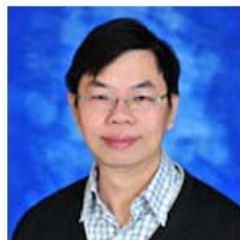
9 - 12 December 2018, Taichung, Taiwan

vcip2018.org

General Chairs



Wan-Chi Siu
Hong Kong Polytechnic University



Chia-Wen Lin
National Tsinghua University

Program Chairs



Wen-Huang Cheng
Academia Sinica



Gene Cheung
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Remarks



Visual Intelligence

Big Data + AI



Let's exchange ideas!



Wen-Huang Cheng



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